3D Gaussian Splatting: A Comprehensive Survey of Representations, Algorithms, and Emerging Frontiers

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Contents

1 Introduction

The evolution of three-dimensional scene representation constitutes a foundational narrative in computer graphics and vision, one characterized by a persistent tension among rendering speed, visual fidelity, and the capacity for scene manipulation. Historically, explicit representations such as point clouds, meshes, and voxel grids offered direct editability and compatibility with traditional graphics pipelines but were limited by discrete sampling and prohibitive memory requirements for complex scenes. The introduction of Neural Radiance Fields (NeRFs) marked a paradigm shift, modeling scenes as implicit, coordinate-based neural networks that map spatial coordinates to color and density values [1, 2]. While this implicit formulation enabled unprecedented photorealistic memory and structural accuracy, it hinged on costly per-ray sampling for volume rendering, severely hampering the ability to achieve real-time functionality [3, 1]. This fundamental limitation created a pressing demand for a methodology that could harness the representational power and subtlety of radiance fields while dispensing with the computational bottleneck of iterative ray marching.

3D Gaussian Splatting (3DGS) has emerged as the field's answer to this challenge, as a technique that bypasses the implicit paradigm by re-introducing an explicit, primitive-based representation. The core of this method is the use of anisotropic 3D Gaussian kernels defined by position, covariance, opacity, and color attributes represented in spherical harmonic coefficients, resulting in a continuous and differentiable scene representation [2]. The key to its breakthrough is a tile-based differentiable rasterizer [4]. A key achievement of the original 3D-GS method is the replace the slow, per-pixel ray sampling typical of NeRF with a highly parallel, tile-based rasterization approach. This allows the splats to be projected into image space, sorted by depth, and alpha-composited efficiently, achieving real-time frame rates without sacrificing the quality inherent to a volumetric representation [5]. This shift from implicit neural fields to explicit point-based primitives re-establishes the advantages of editability and structure, which were largely lost in coordinate-based models, and directly addresses the critical trade-offs among rendering speed, visual fidelity, and editability that have motivated this field's evolution [1, 1].

The conceptual power of 3DGS extends far beyond its original rendering pipeline, demonstrating remarkable adaptability across diverse domains. Its explicit nature facilitates a range of downstream operations that are challenging for implicit representations. The flexibility of the Gaussian primitive allows the use of different data types, e.g., for surface reconstruction, primitive textures, and 3D Gaussian Splatting can be used to increase expressiveness [6, 7, 8], while in physically based modeling, per-primitive materials can decouple geometry from appearance to enable real-time relighting [9, 10]. Advances are also made in rendering speed and storage compression by devising new splatting strategies or high-level parameterized primitives, which either speed up the rendering process or the structure improvements of the original model [11, 12, 13]. The drive towards scalability has led to hierarchical 3DGS representations and the development of compressed representations to handle extremely large datasets [14, 15].

The trajectory of 3DGS is one of rapid evolution, initially as a more creative view-synthesis tool and becoming a versatile foundation for a broad range of research directions and application areas. Its capacity for explicit scene understanding is visible in modern methods for semantic segmentation and feature embedding, while in robotics and autonomous navigation, it provides a direct link between photorealistic reconstruction and geometric composition, serving as a robust map representation [16, 17, 18]. Moreover, dynamic environments tailored to the spatial-temporal domain and reconstruct dynamic scenes by compressing data at the spatial-temporal level, beyond the static case [19, 1]. Given this extensive and rapidly expanding body of work, it is essential to provide a structured overview that synthesizes the advances of 3DGS over time. This survey offers a comprehensive and principled reference of the field, and systematically examining the mathematical foundations, rendering pipelines, and extensibility of 3DGS across different applications and scenarios. We layout the theoretical core, discuss the key aspects that have strengthened its robustness, and analyze how it has been adapted to 3D dynamic scene reconstruction, scalability, and editing. We also outline the trajectory of future research and identify the remaining obstacles, offering a roadmap for the maturation of this field.

2 Mathematical Foundations, Rendering Pipeline, and Optimization

2.1 Mathematical Formulation of Anisotropic Gaussian Primitives

The foundational representation in 3D Gaussian Splatting (3DGS) is a collection of anisotropic, differentiable volumetric primitives that collectively define a continuous scene function. Each primitive is explicitly parameterized by a center position vector μ, a full 3D covariance matrix Σ that encodes its anisotropic shape and orientation, a per-primitive opacity α, and a set of coefficients for view-dependent appearance, typically spherical harmonics (SHs) [2, 1]. The covariance matrix is the geometric core of the primitive; it is factored as $\Sigma = \mathrm{R}\ \mathrm{S}\ \mathrm{S}^T\ \mathrm{R}^T$, where S is a diagonal scale matrix and R is a rotation matrix. This factorization guarantees that Σ remains positive semi-definite and is optimized through a gradient-descent-friendly unconstrained decomposition [5, 20]. The color information is anchored to the primitive through SH functions defined over the unit sphere of the view direction; this yields a view-dependent radiance field that is differentiable and can approximate diffuse through moderately specular appearances. While this parameterization has proven highly effective, recent analyses suggest the SH basis is not inherently taylored to the physical scattering behaviour of many real-world materials, particularly in the context of strong reflective or specular components, which may cause coarse or anisotropic artifacts [10, 9]. This has motivated the exploration of alternative appearance models, such as bidirectional scattering representation functions, spatial varying texture maps, and per-primitive anisotropic spherical Gaussians, allowing more physically grounded decomposition of diffuse and specular terms [10, 7]. This shift towards decoupling appearance from geometry is a recurring theme in recent literature, with works exploring textures as an efficient carrier of fine-grained details while reducing the need for primitive proliferation [7, 21, 8].

A more fundamental set of extensions targets the functional mapping used to describe the primitive density itself. The original GS formulation conceptualizes each primitive as a 3D Gaussian ellipsoid; althought elegant, this hinders precise surface reconstruction because surface-distributed opacity overlaps are reconstructed independently. To better align the radiance field with the geometry, recent work now makes use of 2D oriented planar Gaussian disks, which can act as surfaces [6, 7]. Such implicit plans yield an effective model of surface continuity and consistency but are less capable for volumetric anisotropic measurement. More recent work, constructing a generalized mathematical primitive, extends this to 3D arbitrary kernels as they are parameterized by Fourier descriptors, yielding an inherent scalability in terms of per-primitive fidelity, which is a line of reasoning that is now being tested with surfaces especially for adaptive quality retention [22]. Highly expressive primitives, such as a spectral network based point representation, are also being developed as a way to learn each point's shape and appearance field [21, 21]. These "neural point splats" are compliant in decomposition by having each wavelet have a deterministic analytical kernel but with a neural index for appearance variation [21]. A significant concern is how per-primitive variance and directional radiance is evaluated in practice, and the 3D Gaussian representation is still composed of a set of unequal parameters that adopt the kernel basis and other geometric parameters for primitives. The basic 3D Gaussian representation, however, is still composed of a set of unequal parameters and is the foundation that most of the line of work is built upon.

2.2 The Differentiable Rasterization and Volume Rendering Pipeline

The rendering pipeline of 3D Gaussian Splatting (3DGS) constitutes the algorithmic bridge between the continuous, volumetric scene representation and the discrete image observed by a virtual camera. This stage must reconcile three competing imperatives: the integration of the Gaussian field along the viewing ray, the real-time performance demands of modern graphics systems, and the strict differentiability required for end-to-end photometric optimization. The canonical solution in the original 3DGS framework achieves this through a three-stage pipeline that mirrors the structure of traditional GPU-based rasterization, yet departs from it in ways that manage the unique challenges of overlapping, semitransparent primitives. This success has subsequently spurred diverse architectural modifications to address the structural and approximate bottlenecks of the initial algorithm (e.g., efficient rasterization, order independence). As we will see, these rendering-level decisions are not merely implementation details; they deeply constrain how primitives interact, which in turn shapes the subsequent optimization and density control strategies discussed in the following section. Conversely, the expressiveness of the primitive representation described in the previous subsection—whether anisotropic ellipsoids, planar surfels, or neural-point primitives—directly determines the viability and complexity of the projection and compositing stages that we now dissect.

Projection and Splatting of Gaussian Kernels. The first stage transforms Gaussian primitives from world space into the desired screen space. The correct projective mapping of a 3D Gaussian parameterized by a covariance matrix, Σ, onto the image plane is a homography, specifically an affine approximation along the viewing ray. We posit the camera transformation by a viewing matrix that is composed of a world-to-camera Euclidean transformation $\mathbf{W}$ and a perspective projection with an affine approximation at the point of the Gaussian's center [11]. This yields the 2D covariance matrix $\Sigma' = \mathbf{J}\,\mathbf{W}\,\Sigma\,\mathbf{W}^\intercal\,\mathbf{J}^\intercal$; where $\mathbf{J}$ is the Jacobian part of the locally affine approximation of the projective transformation. Despite this projective approximation being one of the primary sources of error in 3DGS rendering, this step maps the original 3D kernel into an equivalent 2D splat, creating a rasterization-friendly primitive. Directly rasterizing Gaussian kernels in full 3D—which would be equivalent to performing volume ray-casting—is serially expensive due to the per-pixel composition of overlapping kernels across the ray. This orthographic-projective approximation circumvents this by flattening the kernels at a per-Gaussian level rather than per-pixel. However, this approach is inherently error-prone for high-depth-contrast scenes, for which the local affine projection assumption is poor. Recent work has therefore sought to improve the physical grounding of this abstraction— for instance, by treating the Gaussian not as a spatial kernel but as a stochastic solid in 3D, where the rendered radiance for a pixel is integrated along the ray rather than by evaluating planar projections [23, 24].

2.3 Tile-Based Primitive Sorting and Alpha-Compositing

Once the 2D Gaussian footprints are computed, the pipeline organizes the screen into non-overlapping 16×16 pixel tiles. Per tile, each Gaussian must be retained if its footprint intersects the tile. This is the crucial step that bounds the per-pixel workload to a few dozen intersecting kernels per tile, rather than the total number of Gaussians in the scene. Next, the sort is performed along the depth axis within each tile by projecting the position of the Gaussian's center and taking its depth z along the camera ray. This generates a per-pixel ordered list of primitives. Front-to-back compositing then proceeds term by term, accumulating color and opacity. The final pixel color $\mathbf{C}$ is computed via the effectively discretized classical volume rendering integral:

$$\mathbf{C} = \sum_{i \in \mathcal{N}} \mathbf{T}_i\, \alpha_i \mathbf{c}_i,$$

where α_i and $\mathbf{c}_i$ are the opacity and color transmitted by the i-th Gaussian, and transmittance $T_i = \prod_{j=1}^{i-1}(1 - \alpha_j)$ is the cumulative product of opacities encountered so far along the ray. Significantly, the opacity α is not a simple constant per primitive; it is the value of the 2D Gaussian footprint as evaluated at the pixel center, multiplied by the per-Gaussian learned opacity, and potentially modulated by an "alpha texture" (as in textured Gaussians) for finer spatial control of transparency [25, 26, 13]. This “volume-rendering-like” weighting turns the rasterization into a differentiable and trainable approximation to volumetric integration, bridging the gap between point-based graphics and neural volumetric rendering [23, 7].

2.4 Alternatives to Explicit Depth Sorting

The deterministic per-tile depth-sorted compositing of the classical 3DGS pipeline carries two inherent limitations: (a) it enforces a single global ordering of Gaussians along each ray, which is insufficient when overlapping, semitransparent, or intersecting structures have no meaningful depth ordering; (b) the repeated sorting of per-slice lists per pose adds a significant computational overhead. These order-dependence assumptions cause transparency artifacts when the kernels share nearly identical depth along the ray.

This motivates a line of work that seeks order-independent transmittance computation or alternative sampling strategies. A notable reformulation adopts *surfel* representations and computes exact ray-splat intersections, instead of approximating the 2D footprint and promoting global depth order—see for instance the *2D Gaussian Splatting for Geometrically Accurate Radiance Fields* [6]. In contrast, *Splat the Net* [21] bypasses the creation of a per-pixel RGBA list altogether, using globally queried neural primitives that are continuous and layered at render time—sidestepping depth-list enumeration at the cost of altering the rasterization paradigm. Meanwhile, recent works attempt to accelerate or replace the initial deterministic depth sorting by incorporating Markov Chain Monte Carlo (MCMC) sampling to select relevant Gaussians [27], improving physical accuracy without a proportional loss of speed. More generally, alternatives to 2D depth-sorting can achieve higher physically correct transparency and multi-layer compositing but often pay for this with increased memory or computational overhead—the central trade-off in this line of developments.

2.5 A Co-Design View and Future Projection

Beyond the global pipeline, the success of the original tile-based splatting relies on the coordination of all operations onto GPU-accelerated, massively parallel hardware. However, the rendering model is increasingly approximating the continuous volumetric integration, which limits fidelity and introduces potential artifacts. The recent development of 2DGS-style work, with exact ray-splat intersection geometry and per-ray transparency, points toward surfel-aware models but at the cost of more complex per-ray computation [6]. Similarly, stochastic solid formulations and Gaussian surfel methods push fidelity closer to true volume integration while retaining a slice of the computational advantage [23, 27]. These innovations mark a relocation in the design space from the depth-sorted projection toward physical plausibility, albeit with potential performance overheads.

Recent developments have also bridged the complementary strengths of mesh and Gaussian primitives, particularly for high-quality free-viewpoint video rendering [28]. Moreover, the efficient and stable optimization of primitive states benefits from geometry-guided initialization and dynamic density control methods [29]. Such developments point toward co-design trajectories where rendering, density control, and optimization are increasingly intertwined. As the adaptive density control (ADC) heuristics and loss functions discussed in the following section evolve toward closed-loop, statistically principled strategies, the rendering quality achievable per primitive—and thus the fidelity limits of the rasterization stage itself—will be further refined.

In summary, the rendering pipeline has evolved from a hard constraint of memory-bound to flexible approximations of the physical volume without sacrificing differentiability, marking a path toward more faithful and flexible implicit rendering while retaining real-time speed. This co-design of optimization and temporal-streaming, together with the robust initialization therapies analyzed in the next section, will drive the next step of the dominant method.

2.6 Optimization, Loss Design, and Adaptive Density Control

The optimization of 3D Gaussian Splatting (3DGS) constitutes a delicate orchestration of photometric objectives, geometric heuristics, and adaptive population control. At its core, the training objective combines an L1 pixel-wise error with a structural dissimilarity term (D-SSIM), yielding the composite loss L = $(1 - \lambda)L_1 + \lambda$L_{D-SSIM}, with λ typically set to 0.2. This joint formulation leverages the complementary strengths of per-pixel fidelity and perceptually-aware structural alignment, yet its uniform application across the image introduces an inherent inefficiency: all regions are penalized equally, regardless of their reconstruction status. Recent analyses suggest that this operation acts as a smoothing mechanism over gradients from diverse viewpoints, impairing the discovery and refinement of geometrically defective regions. Indeed, the average aggregation of view-space positional gradients can obscure the contributions of a small subset of views, particularly where errors are spatially concentrated [30]. This insight has motivated a shift toward loss functions that are explicitly sparse—masked to focus optimization on under-reconstructed regions or where per-pixel rendering errors exceed a predefined threshold. By amplifying gradients from significant, outlier-driven differences, these targeted losses allow refined Gaussian growth and mitigate the "needle-like" artifacts that arise from uniform supervision [30].

Central to the parameter refinement and topology evolution is the adaptive density control (ADC) framework—a heuristic set of operations aimed at rectifying both under- and over-reconstruction. The original ADC algorithm relies on the accumulated view-space positional gradients of each Gaussian. A primitive is split if the average magnitude of its gradients exceeds a threshold and its scale decays below some scale threshold, while it is cloned when gradients remain high but scale is large. Under extreme under-reconstruction, pruning removes Gaussians with opacity below a minimum threshold. While computationally simple, this pair of thresholds, guided by mean gradients, suffers from several vulnerabilities: it fails to differentiate between Gaussians that consistently accumulate moderate gradients from those that spike sharply from a single direction, and it induces a slow growth dynamic prone to equilibrium states of redundant primitives. Furthermore, this heuristic can allocate unnecessarily dense clusters of small Gaussians on smooth surfaces, while sparsely populating those with high-frequency detail [31, 32]. The default heuristic's reliance on mean magnitude as a signal for deficiency also mistakes the ghosting of volumetric geometry for genuine detail, occasionally producing artifacts in underconstrained or over-fit scenarios.

Consequently, a substantial body of subsequent research has reformulated the densification and control process as statistically principled, error-driven, or representation-aware rather than a purely heuristic approach. For instance, error-driven densification directly correlates rendering error maps to new primitive placement, explicitly tackling the regions that cause high loss with new primitives and overlooking well-reconstructed areas [30]. Alongside this, direction-aware growth checks multi-axial statistics of gradients, distributing clones or splits not just at region thresholds but along the axes where a primitive's footprint misaligns with the geometry. Primitive-count-efficient variants further highlight a key limitation: the standard Gaussian scaling-free opacity renders anisotropic features inefficiently, requiring additional attention to the opacity representation to achieve scene fidelity [32]. Similarly, Deformable Radial Kernel Splatting (DRK) presents a flexible primitive with adjustable boundaries, and shows better reconstruction quality with reduced primitive counts [31]. A key observation emerges: the densification metric changes with the primitives' shapes; when using primitives with more complex boundaries, the placement and redundancy metrics become inseparable from the shapes being added [Deformable Radial Kernel Splatting;

Fourier Splatting]. An over-generalized representation can impede effective densification—spatial sampling must be aligned with the representation's expressiveness to build an efficient hierarchy (§2.1).

To stabilize and compactify these distributions, modern ADC mechanisms have diverged into distinct camps: opacity-controlled methods (opacity with view-dependent channels), entropy regularizers, error-driven placement, and probabilistic or series-based priors. In the domain of opacity, the sigma of a Gaussian can serve as a decisive **filter**: for instance, opaque, hidden, or dispersive splats receive low opacity and can be pruned to maintain accuracy. In terms of rendering quality, adjustments in the alpha computation, such as analytic integration with anti-aliasing, ensure that pixel footprints are considered rather than treating pixels as isolated points of sampling [33]. This redesign improves the anti-aliasing capability of the model with superior detail [33]. Similarly, the Gaussian "Blending" soft-formulation replaces alpha blending with a spatially-averaged transmittance distribution, addressing issues that arise from deterministic view sampling [34, 35]. Notably, the recent work brings order-independent transparency alternatives: by decreasing opacity thresholds, rendering can leverage of culling and pruning to accelerate the pipeline [36]. In contrast, in scenarios where sorting is eliminated from the pipeline, Gaussian opacity estimates are replaced with spatial proxies—such as cell-mapped chips or *contributions-aware* proxies—that preserve final compositing accuracy [37, 38, 39].

Where these methods differentiate is **another** central tension: whether to view adaptive density control as weight-space-based or update the **canonical** effective as a MCMC-style hierarchical. Principled MCMC-based methods, that consider adding and pruning Gaussian positions via updates sampled to match an underlying distribution within the scene [40], remain a mathematical consideration rather than a widely used technique. In contrast, the **Entropy Regularized** methods compute a scalar statistic on the Gaussians as a "contribution score," and enforce smooth reduction based on quantile comparisons [41]. One notable vector is **specular Gaussian**: using a spatially variable (view-dependent) opacity encoding with anisotropic channels that do not require shrinking the number; these distributions encode more details and need fewer Gaussians to represent high-frequency, specular behavior [42]. The trend is toward attributing more *representational* capacity to each Gaussian *including* opacity design ("view-dependent opacity") and allowing fewer to represent a scene—point not only a computational advantage but also a path to shape-transparency decoupling; this is contributing to both refactor.

Nevertheless, the common algorithmic architecture still treats "densification → inheritance → pruning" as coarse-grained periods of training with distinct modes; however, the **acceleration** of these processes is only recently being integrated from preprocess. The cascade is not time-conceding: recent hardware-accelerated-aware approaches, such as tile-based culling, warp-coherence, or performing part of the culling in an earlier preprocess stage, actively create positive feedback loops with the densities and pool they reproduce [43, 44, 45]. This coupling <concludes> is what enables the substantial speedups observed (e.g., 1.33–1.88× for tile-based CSP, 7.76× Deep Blending) while preserving accuracy [46, 35, 45]. More broadly, recurrence-aware surrogate networks for transmittance estimates enable *mixed* processing on low-power hardware, indicating that the future of ADC could be gradient-less and caching-based [38, 47].

Looking to the future, there remain key open problems: (i) the cascade of "densification → pruning → relaxation" training is staged and rarely formulated with closure; (ii) mitigating floaters and redshift is effectively linked to high-level transparency priors that simultaneous affect both loss and topology; (iii) the single-pass, fixed-end performance-artifact tradeoff is avoided— [48]. Recent advances in continuous **regularizers** for opacity and density (e.g., *isotropic* Kernels, geometry-like normal constraints) could offer a pathway [49, 50]. Indeed, there is capacity in "4D" representations and in *fine-scale* mesh-surface contact that will push current analog limits [50, 51].]. However, on the immediate horizon, it seems likely that researchers will keep the **two-track** heuristic (loss-based, stat-driven) at their core, injecting robustness and efficiency where possible. The most obvious path, though, is to formulate \$3DGS\$ *as a sampler and integrator*: treat loss+ density + culling as a single, unified operation, and save polygon-level accuracy for systems that can afford the full neural or volumetric renderer [52]. This co-design of optimization and temporal-streaming is exactly where the next steps of the dominant method will occur, I argue.

2.7 Initialization Strategies and Robust Optimization from Suboptimal Inputs

The fidelity of 3D Gaussian Splatting reconstruction is fundamentally contingent upon the initialization of its primitives. The canonical pipeline leverages Structure-from-Motion (SfM) outputs to seed the optimization. While this provides a geometrically plausible starting distribution, the correspondence-based nature of SfM introduces inherent fragility. In textureless, repetitive, or sparsely observed environments, the initial point cloud is often sparse, biased towards high-contrast features, and can even be absent in critical areas, leading to optimization failures or slow convergence. Moreover, the sensitivity to the initialization extends beyond mere point placement; the subsequent Adaptive Density Control (ADC) process, which heuristically clones, splits, and prunes primitives based on positional gradients, is critically dependent on this initial configuration to guide the gradient flow towards an accurate solution.

Recognizing that accurate initial seeds are a luxury not always available, recent research has focused on making the optimization robust to suboptimal or unreliable inputs. A primary axis of this research targets the refinement of the density control mechanism itself, shifting from rigid heuristics to more principled, error-driven policies. Given the fragility of gradient-based densification, several works propose a pivot from positional-gradient heuristics to more spatially explicit signals. For instance, a pixel-error-driven formulation leverages a per-pixel error map to guide densification, correcting underlying biases while also controlling the total primitive count [53]. The introduction of a directional consistency (DC) constraint on positional gradients, which uses the angular coherence of gradients rather than solely their magnitude, improves process reproducibility, and eliminates redundant splitting in regions that can be represented by off-axis sub-primitives [54]. Similarly, drawing on the analytical insight that the standard ADC heuristic is also prone to generating redundant points, a framework based on the gradient of opacity can be adopted as a lightweight, yet robust, proxy for rendering error, rather than standard position-based estimates—showing a major improvement in few-shot settings by preserving model compactness and countering the formation of floaters [55].

Beyond the criterion for densification, the strategy for merging and eliminating redundant primitives has become a focal point for robustness. To avoid the destructive optimization cycles that can limit the exploration of the loss landscape, several sophisticated integration mechanisms have been proposed. Locality-Aware Density Control (LocoADC) addresses the inefficiency of the original algorithm by jointly analyzing the local continuity of reconstruction errors in image space and the similarity of neighboring Gaussians in parameter space; this dual-locality principle enables the clustered densification of under-reconstructed content and the merging of over-reconstructed, redundant primitives, leading to a more compact and accurate representation [56]. Complementary to this, density control can be interpreted through an optimization-theoretic lens, where densification is framed as a strategy to escape from suboptimal optima in the geometric loss landscape, leading to works that propose a steepest descent strategy to minimize the number of clones for a given update direction [57]. This theoretically motivated approach can yield approximately a 50% reduction in point primitives without sacrificing rendering quality. The critical implication is that effective, robust optimization does not require more primitives, but rather smarter and more accurately parameterized updates that emulate first-order optimization on a continuous space.

A different vein of work introduces a paradigm shift in density control by making the control mechanism itself learnable. Rather than relying on handcrafted schedules, these methods formulate densification and pruning decisions as a parameterized policy network trained via Reinforcement Learning to decide when and where to adapt the primitive population [58]. This subtle shift allows the model to reason about scene geometry in a way that is well-posed across scenes. To make such a framework computationally tractable—a primary hurdle for any learning-in-the-loop approach—a closed-form, high-level sensitivity analysis has been derived that reduces the naive $O(N^2)$ calculation to $O(N)$, turning the process from an empirical workaround into a practical algorithmic component [58]. This principle is central: without a proper and efficient management of the reward stream, the overhead of the secondary optimization would likely impede the overall performance of the method.

In contrast, some methods focus on the overall training dynamics. The density of control. To achieve robustness, explicit modifications to the scaffolding can be made via a *target point control* schema. The optimization landscape is full of global and local minima depending on network initialization and training in the standard formulation. To avoid entrapment, the target point control uses a quota-like mechanism (a quadratic controller for final point counts) that evenly adjusts the densification and opacity-culling hyper-parameters themselves, and this controller is robust *independently* of the point distribution during training, as long as a controlled number of total iterations within the densification window are enforced [59]. This is an important lesson: the iteration count at which the density variations are applied modifies strongly how the final result depends on the initialization, and by fixing a *mid-term* cap (target), we can effectively improve algorithmic stability by ensuring a more uniform error distribution across the view space, allowing fairer comparisons and baselines for all methods.

Looking ahead, the synthesis of these approaches suggests a step aside from the simple reliance on the initial SfM seed and the naive gradient-based refinement. A more balanced (efficient) approach emerges by breaking the robustness into two integrated axes: *control-space* robustness and *optimization-space* robustness. The former includes modifications to the density criterion (for replacing the heuristic), the shape for correct gradients, and the merging for redundancy. Simultaneously, an important future direction calls for the unification of these often ad hoc control mechanisms under a differentiable, robust, and efficient objective, learning the adaptive dynamics for the whole process to drive quality from the first stage. With neural networks or explicit optimization, this practical refinement allows us to amortize the best adaptive profile across scenes. The step from fixed reflex actions to model-driven decisions is the difference between bridging heuristics and true robustness from suboptimal inputs. Such controlled and learned generation schemes will also serve as a natural bridge to the speed-oriented, loss-prioritized techniques introduced next, as refinement and acceleration become increasingly symbiotic.

2.8 Acceleration and System-Level Optimization

The optimization of 3D Gaussian Splatting (3DGS) models is frequently constrained not by algorithmic convergence alone but by the substantial computational and memory overhead inherent in both the training and rendering phases. This subsection examines the algorithmic strategies and system-level innovations designed to accelerate 3DGS, focusing on mitigations for memory bottlenecks, enhancements in parallel efficiency, and the reduction of the model's overall computational footprint.

At the algorithmic level, a primary direction accelerates training by recognizing that not all pixels contribute equally to the optimization. Hard-pixel sampling methods identify and prioritize under-reconstructed regions during the loss computation, thereby skipping well-converged areas and focusing computational effort on informative, spatially structured regions that reduce wasteful iterations [60]. This region-prioritized optimization, which couples structural awareness with saliency-based weighting, is shown to accelerate convergence by up to $3\times$ while simultaneously reducing the Gaussian count by up to $4\times$ [60]. The reduction in model capacity is also tackled at the parameter level. Spatially structured pruning and compact representation methods directly address the memory footprint and computational redundancy of a trained model, offering compression by eliminating less significant primitives and enforcing spatial regularity without a substantial loss in fidelity [60]. Furthermore, the standard reliance on heuristic densification has been identified as a significant bottleneck; methods like EDGS propose a one-step approximation of scene geometry via triangulated dense correspondences, which eliminates the need for iterative densification and parallelizes the initialization of all primitives, dramatically shortening the optimization path and improving both efficiency and high-frequency detail restoration [32]. This idea is further extended by learned densifier networks that predict new Gaussian placement based on rendering gradients in a single update, removing the dependence on hand-crafted densification heuristics and improving the geometric fidelity of flat wall structures, accelerating runtime by up to 8× [61].

Beyond algorithmic improvements, the orchestration of data across the GPU hierarchy is critical for system-level performance. The sequential dependency introduced by incremental densification—where new Gaussians must wait for updates—creates significant latency. Progressive or on-the-fly frameworks tackle this challenge with a block-based data flow. By updating new images and their overlapping images in local and semi-global stages and using an adaptive learning-rate schedule, these systems can shorten the time required to incorporate new images into the existing model, thereby achieving near real-time optimization [62]. Furthermore, to address memory-bound scenarios with millions of splats, out-of-core systems rely on splitting the scene into manageable blocks (e.g., via KD-trees) and hierarchically mapping the Gaussian data to control the creation of intermediate data and the scheduling process on GPUs [62]. This hierarchical mapping allows only relevant subsets of the model to be streamed to the GPU for rendering.

Hardware-accelerated and tile-level optimizations form another crucial layer. Beyond simple GPU usage, specialized implementations target the distinct workload profile of the splatting algorithm. This includes the use of split-voxel hierarchical mappings and key-voxel-based acceleration to avoid computing the full multiply-accumulate operation for every Gaussian-anchor interaction, using axis-shared computational units instead [63]. On the rendering side, the system is structured to avoid overfitting and accelerate convergence by redesigning the Gaussian optimization itself, decoupling static and dynamic scene components, and using a delayed growth strategy to reduce redundant computations [38]. Moreover, the optimization landscape itself can be modified to favor easier convergence. The "blur trap" analysis reveals that vanilla 3DGS optimization is biased toward local suboptimal solutions (blurriness) due to the gradient patterns of unexplored regions. This has been addressed by introducing simple explicit exploration strategies—random Gaussian seeding and vertical splitting—that push the optimization out of these traps, achieving convergence with fewer iterations [64].

Finally, scalability to ultra-high-resolution and distributed environments presents a frontier for these optimizations. While point-wise rasterization is inherently parallelizable, it does not inherently distribute across multiple devices. Advanced partitioning and visibility-aware scheduling frameworks distribute the workload across multiple GPUs or nodes, decomposing the Gaussian model and retaining near-real-time frame rates. For scalable processing, learned initialization from a 3D foundation model enables a reconstruction speed that produces a ready-to-use Gaussian model in approximately three minutes without preprocessing, marking a step change in what constitutes "real-time" [63]; this is a clear convergence of speed and scale. Collectively, these advancements demonstrate that continuous acceleration is being achieved by a symbiosis of re-designed core optimization logic, data flow, and hardware utilization, rather than purely by scaling compute resources.

3 Quality Enhancement, Regularization, and Robustness

3.1 Scale-Aware Rendering and Anti-Aliasing

The discrete, primitive-based nature of 3D Gaussian Splatting (3DGS) introduces a fundamental challenge: the representation of a continuous scene is inherently tied to a specific sampling rate, determined by the training data and the spatial density of primitives. When rendering at resolutions or viewing distances that deviate from this training distribution, the model becomes susceptible to severe artifacts. Two principal failure modes are observed. Zooming in exacerbates the *dilation* artifact, where the splatted Gaussian kernels, projected to larger screen-space footprints, result in a blurred, over-smoothed appearance. Conversely, zooming out triggers *aliasing*, a consequence of the high-frequency content encoded in the spherical harmonics and primitive positions being undersampled at lower resolutions, leading to flickering and shimmering across frames. These issues are not mere cosmetic flaws but represent a fundamental breakdown of the scale-equivariance of the representation, hindering practical deployment in applications demanding visual consistency across varied focal lengths and distances [1, 2].

A pioneering solution to these artifacts is the **Mip-Splatting** approach [1, 2], which identifies the root causes as the unbounded frequency content of 3D Gaussians and the resolution-dependent 2D dilation filter. To mitigate the former, it introduces a 3D smoothing filter that constrains the primitives' frequency content relative to the training sampling rate, effectively eliminating high-frequency peaks that cause aliasing at long distances. To address the latter, it replaces the 2D dilation operation with a 2D Mip filter, whose size is modulated by the current sampling rate. This prevents the aliasing and dilation when zooming out, enabling multi-scale consistency by conditioning the effective kernel size on the image resolution and camera distance. This frequency-conscious filtering concept can be further reinforced through physical regularizations, such as multi-scale density control or Pyramidal Multi-Scale Supervision (PMS), which enforces consistency across an image pyramid. By supervising the densification and primitive evolution with data at multiple scales, these strategies prevent the optimization from converging to degenerate, jagged primitives that only appear correct at a single scale [20].

While Mip-Splatting treats the scale problem as a kernel filtering issue, an alternative strategy is to manage the complexity of the scene representation directly via **Level-of-Detail (LOD) and hierarchical structures**. In the context of 3DGS, this involves creating a hierarchy of primitives that may be selected by the renderer based on viewing conditions, ensuring consistent rendering speeds and quality across a vast range of scales. Octree-GS introduces an LOD-structured 3D Gaussian approach, where multi-resolution anchor points and hierarchically organized Gaussians are dynamically selected for rendering. This not only accelerates computation by avoiding the rendering of unnecessary primitives for distant content but also provides a principled solution for very large, complex scenes [65]. This philosophy mirrors the hierarchical representation and divide-and-conquer strategy integral to rendering very large datasets, where scenes are represented with a hierarchy that maintains visual quality while enabling efficient Level-of-Detail selection [14]. For the ultimate scalability, a hierarchical organization is vital, ensuring consistent rendering performance and resource adaptation for massive scene collections.

A more recent direction explores rendering at arbitrarily, user-defined scales without requiring changes to the underlying model, sometimes termed non-linear or super-resolution rendering [20]. This is often achieved by re-parameterizing the primitive's mathematical formulation. For instance, the Fourier Splatting approach [66] uses scalable primitives with Fourier-encoded descriptors. By truncating the Fourier coefficients at query time, it enables the inherent scalability of the model to control the level of detail, avoiding the need to prune elements. These arbitrary-scale methods implicitly address aliasing by explicitly augmenting the model with the parameterisation from the target output and often leverage generative priors to ensure structurally consistent views.

In summary, the area of scale-aware rendering is converging on a combination of filtering, representation, and supervision strategies. Mip-style filtering offers a principled solution for a continuous spectrum of scales, while LOD hierarchies provide an efficient alternative for large-scale browsing, often relying on discrete levels rather than continuous ones. Future research will likely focus on the integration of these strategies with real-time architectures, as well as on frequency-decoupled appearance (beyond spherical harmonics) to support truly scale-invariant world models. This will be vital for practical applications, especially in areas like automotive viewing and SLAM, where focal length and viewing distance vary rapidly and continuously [67, 68].

3.2 Geometric and Surface Regularization for Accurate Reconstruction

The optimization of 3D Gaussian Splatting (3DGS) is fundamentally driven by photometric loss functions, which, while effective for view synthesis, remain agnostic to the underlying physical surface. This loose coupling between the discrete radiance primitives and the continuous scene geometry frequently results in characteristic artifacts such as floaters, holes, and noisy or non-manifold reconstructed surfaces. Addressing this, the research community has developed a suite of geometric regularizations, representation adaptations, and optimization strategies, broadly categorized into three fronts: normal-guided depth regularization, surface-aligned primitive transformations, and targeted optimizations to escape geometric local minima.

An early prominent line of work directly incorporates photometric and geometric consistency constraints into the optimization loop. For instance, a general multiview geometric regularization integrates depth and normal priors from Multi-View Stereo (MVS) into the GS pipeline, using a median-depth-based relative loss with uncertainty to fuse MVS depth with the GS-optimized positions, effectively capitalizing on their complementary strengths in regions of varying color and texture [69]. This approach exemplifies a shift from purely appearance-based objectives to hybrid signals that ground the optimization in geometric space. However, such regularization strategies rely on the availability of sufficiently reliable external priors, and their effectiveness diminishes in settings where those priors themselves are noisy—a limitation that becomes especially critical in sparse-view regimes discussed in the next section. Other approaches have sought to guide optimization by providing a more structured initial state. Two key limitations of the standard optimization—its dependence on accurate initialization and the tendency for unstructured Gaussian distributions to converge on unordered surfaces—are addressed by geometry-guided initialization and surface-aligned optimization strategies [29]. These methods predict initial parameters to ensure precise placement and update them in a way that aligns with the surface normals for improved shape reconstruction. These systematic issues are further compounded by geometric local minima inherent to the optimization landscape, where the Gaussians converge to physically inconsistent configurations despite low photometric error.

This pivotal insight—that the shape and orientation of the primitive itself is a significant source of geometric ambiguity—drove the development of surface-aligned primitives. This has motivated the shift forward from volumetric 3D kernels that represent volume fills to planar 2D disks. By collapsing a 3D volume into a set of 2D oriented planar Gaussian disks, representational power is re-focused on modeling the surface geometry [6]. The broader trend of separating appearance from geometry stabilizes the optimization. A major challenge is the strong coupling between appearance and geometry, which can introduce geometric bias. An illustrative analysis shows that the default 3DGS form struggles to distinctly represent texture and geometry, and a solution introducing an additional per-splat geometric opacity parameter to decouple geometry and appearance was proposed [70]. This simple modification proved effective in complex scenes, including those with transparent objects. In the context of specific primitives, the issue of texture and detail is addressed through per-primitive texturing, which is an effective strategy to decouple appearance from geometry. GStex textures each 2D Gaussian primitive so that even a single primitive is capable of capturing detailed appearance [7]. This distributive philosophy proves more efficient: representing a large diffuse region with a few textured planar primitives is preferable to encoding the same high-frequency appearance into thousands of sub-pixel geometric elements. To reduce geometric artifacts, parameter-efficient primitives that are richer in appearance also inform the design of the texture budget. Adaptive sampling for textured Gaussians and error-driven anisotropic parameterization have been proposed to allocate this budget cost-effectively [71]. This isolates the two goals architecturally rather than relying solely on the optimizer. Several novel forms of primitive geometry emerge from this line of research, such as the Gaussian-Hermite kernel formulation, which introduces a higher-order extension capable of more anisotropic primitives and better capturing object boundaries [72]. Similarly, geometric reconstruction is enhanced by drawing parallels between Gaussian surfels and stochastic opaque surfaces, using them to splat a geometry field—enabling a unified differentiable rendering method [27].

In summary, the field has progressed from simpler regularization (adding geometric depth/normal priors) to the explicit modification of the core representation itself. Moreover, several of the optimization instabilities common in dense-view frameworks are addressed more aggressively in the sparse-view setting, where priors are absent and hallucination must be mitigated through structural redesign—linking the surface-quality concerns discussed here to the regularization and generalization challenges that follow. The future of this line of work lies in the interplay of three dimensions: the decoupling of appearance primitives physically from their geometric counterparts, the development of richer higher-order primitives (e.g., textured disks at anisotropic scales), and the robust adaptive control required to manage these new forms. The synergy of these three enables a key synthesis: a textured geometric representation—where a sparse set of primitives simultaneously explains complex geometry and densely detailed appearance—offers a promising path forward for robust, lightweight, and geometrically sound 3DGS, especially in resource-constrained and high-fidelity scenarios.

3.3 Sparse-View and Generalizable Reconstruction

Sparse-view reconstruction represents one of the most challenging and practically significant frontiers in 3D Gaussian Splatting (3DGS), as the underlying optimization framework was originally designed for dense multi-view inputs with reliable Structure-from-Motion (SfM) initialization. When the number of input views drops to a few dozen or fewer, the reconstruction problem becomes severely ill-posed: photometric losses no longer provide sufficient constraints for the underdetermined optimization, and the adaptive density control mechanism tends to either collapse into degenerate solutions characterized by "blur traps" or produce hallucinated geometry–the well-known floater artifacts. These failure modes are particularly acute in texture-less regions where gradient signals vanish, leaving primitive positions entirely unconstrained. The research addressing this problem is broadly organized along a spectrum of prior strength, ranging from the injection of geometric regularizers and generative diffusion priors to the complete abandonment of per-scene optimization in favor of learned feed-forward reconstruction.

A foundational category of methods **substitutes the missing information by leveraging geometric priors from large-scale pre-trained models**, particularly monocular depth estimators [73, 74]. These methods typically incorporate depth maps as pseudoground-truth supervision during optimization. The depth prior constrains Gaussian positioning along the camera ray, preventing the aforementioned local minima associated with volumetric depth ambiguity. However, the injection of monocular priors is not straightforward—the scale ambiguity between the estimated per-image depths and global scene coordinates must be resolved through alignment strategies. More advanced designs propagate uncertainty estimates into the loss function, applying depth smoothing and normal consistency terms preferentially in regions of low confidence, thereby maintaining flexibility where the estimator is unreliable. Diffusion-model priors represent a more recent extension of this paradigm, providing stronger generative constraints to regularize appearance [73]. By leveraging the strong inductive bias of 2D diffusion models, these approaches align the rendered distribution to plausible natural-image statistics, effectively hallucinating geometry in unobserved regions while ensuring multi-view consistency through the rendering loss of the differentiable pipeline.

A second category **addresses the problem by leveraging dense multi-view structure from matching and view-consistent constraints**. Instead of relying solely on sparse initializations, learning-based MVS models can be deployed to generate dense depth guidance for initializing the Gaussian cloud. This substantially densifies the initialization, providing the optimization with rich geometric signal even where feature extraction fails—a common occurrence in texture-less, low-light, or low-overlap settings. Subsequent research extends this by imposing view-consistent geometric constraints at automatically selected virtual viewpoints.

By rendering novel views and enforcing consistency between the renderer's output and the depth guidance, these approaches introduce a cyclical supervision signal that anchors Gaussians to the scene surface. However, these methods rely on the selection of virtual views: While they effectively mitigate overfitting to the sparse training views, the virtual consistency constraints are often violated in strongly occluded regions and do not fully eliminate texture crawling or floating artifacts.

The most radical departure from the per-scene optimization paradigm, however, is the **feed-forward generalizable reconstruction methods**, which relegate the need for SfM initialization to a deep network. These models, typically built upon Transformers or multi-view stereo (MVS) backbones, consume a set of posed images and regress a complete Gaussian primitive set—positions, covariances, opacities, and spherical harmonics—in a single forward pass [75]. These network architectures capture the global context, allowing them to handle the severe geometric ambiguity of sparse low-overlap views [76]. *Predict-then-refine"* thus emerges as an effective middle ground: the feed-forward predictions provide a significantly stronger optimization prior than random seeds, and a subsequent brief per-scene refinement closes the smaller fidelity gap while retaining robustness. These feed-forward approaches are driving inference to real-time while maintaining robustness to degenerate input, yet the recent hybrid approaches that integrate explicit reconstruction with generative priors for unseen regions point toward the most promising direction for true few-shot fidelity. Sparse-view reconstruction therefore identifies an engineering tension: explicit per-scene optimization may be blended with statistical generalization, and the ability to seamlessly transition between the two modes could define the next generation of 3D reconstructors. While the field is still developing, with existing methods exhibiting degradation when overlaps fall below a threshold, the convergence of geometric guidance and data-driven priors promises a robust pathway toward a truly sparsely-observable Gaussian Splatting.

3.4 Appearance and Degradation in the Wild

While the core 3D Gaussian Splatting (3DGS) framework assumes a static, clean world with perfectly known camera poses, real-world captures invariably violate these ideal conditions. Transient occluders, dynamic lighting, and sensor-induced degradations such as motion blur introduce inconsistencies that the base photometric loss cannot reconcile. Consequently, the reconstruction problem becomes ill-posed in a manner complementary to the sparse-view scenario discussed previously: rather than lacking observations, the optimizer is confronted with contradictory observations that actively mislead the gradient signal, producing floaters, blurred geometry, and ghosting artifacts. This subsection addresses the robust optimization techniques developed to handle such "in-the-wild" captures, categorizing them into strategies that explicitly model and counteract blur, those that suppress transient distractors, and those that jointly refine camera pose and scene parameters.

A dominant source of degradation in casual capture is camera motion blur—an artifact of the camera moving during exposure. Optimizing against blurred images naively forces the optimizer to generate elongated, imprecise Gaussians along the hypothetical motion path in order to explain a spatially integrated measurement, an artifact directly analogous to the "blur trap" local minima observed in sparse-view reconstruction. A principled solution is to model the blur within the rendering process itself rather than leaving it as a byproduct of scene parameters. By integrating the rendering over the camera trajectory during exposure time—either through rigid transformations for shake or continuous deformation fields via neural ODEs for complex camera motion—the optimization is guided to factor the measured blur into a coherent sequence of sharp states. This not only yields a sharp, canonical 3D representation but also effectively deblurs novel views without demanding per-pixel target signals that the sparse-view losses lack. However, this conceptual clarity comes at a computational cost, prompting more efficient alternatives. For instance, a Blur Proposal Network can be trained to predict multi-scale, spatially-varying blur kernels that condition the 3D scene optimization, achieving deblurring at inference speed and avoiding per-input re-optimization.

Beyond temporal degradation, the assumption of scene immutability is frequently contradicted in real-world captures by transient occluders. Pedestrians, vehicles, or transient specular highlights emerge and disappear across viewpoints, and treating them as persistent geometry introduces the same floater artifacts as hallucinated geometry in the sparse-view case, but here they arise from contradictory supervision rather than insufficient supervision. Robust optimization approaches tackle this by learning to assign per-pixel reliability weights or by leveraging semantic segmentations that suppress dynamic distractors. By explicitly down-weighting or masking the contributions of transient elements in the loss, the geometry optimizer is anchored to only the most globally consistent image evidence—parallel to the geometric consistency strategies used in high-quality 3DGS scenes but adapted to operate without explicit prior depth on unconstrained captures. This approach implicit relies on the static background retaining majority signal; the heuristics degrade gracefully only when the transient elements cease to be a minor disturbance.

A third, more systemic category of degradation arises from inaccurately estimated camera poses—a problem ubiquitous in large-scale outdoor scans and low-texture environments where correspondence matching fails. Pose errors create a "chicken-and-egg" optimization loop where scene geometry and camera localization each corrupt the other, exhibiting a pathology parallel to the initialization issues in dense-view reconstruction but amplified by the reality of imperfect calibration. Several modern pipelines address this by decoupling pose estimation, geometry refinement, and renderer updates into separate optimization loops, creating a more stable loss landscape than end-to-end entangled gradient descent. More recent trends are moving away from per-scene monolithic optimization for camera estimation altogether, using test-time refinement and latency-aware algorithms that are robust to pose noise without requiring explicit re-optimization of parameters.

In summary, handling the uncontrolled—and often genuinely messy—nature of real-world input necessitates decoupling the problem beyond a simple reconstruction objective. The state of the art blends explicit physical priors—camera motion models, blur kernels, semantic segmentation, and pose refiner—with learned algorithms to robustify the optimization. Much like the sparse-view case, the core design principle is to separate blessings of reliable evidence from sources of contamination while allowing learned priors to fill the gaps left by unreliable measurements. Future work will increasingly integrate these robustification strategies into end-to-end systems, closing the gap between the controlled lab environment and truly dynamic, unconstrained deployment settings.

4 Scene Representation, Geometry Extraction, and Compression for Scalability

4.1 Surface Extraction and Geometry Reconstruction

The transition from view-synthesis-oriented radiance fields to explicit, surface-aware geometry represents one of the most consequential evolutions in 3D Gaussian Splatting research. The native 3D Gaussian primitive—an anisotropic ellipsoid with view-dependent appearance—is optimized purely for photometric fidelity, leaving its spatial distribution only loosely coupled to the underlying surface [1, 2]. This decoupling manifests in the well-documented "floater" artifacts, where semi-transparent Gaussian shells aggregate to satisfy photometric losses across inconsistent viewpoints. While the explicit, point-based nature of the representation offers a natural starting point for surface extraction—similar to classical point cloud reconstruction—naively extracting geometry from Gaussian centers yields noisy, hole-ridden meshes with poor surface alignment [77].

To address this fundamental misalignment, a major research thrust has focused on adapting the primitive itself. The most influential development in this direction is the introduction of the 2D oriented planar Gaussian disk [6], which collapses the 3D volumetric ellipsoid into a flat disk with a normal orientation. This perspective-accurate 2D splatting technique provides view-consistent geometry that intrinsically models surfaces rather than volumetric blobs. By discarding the third scale dimension, 2DGS enforces a strict planar surface prior at each primitive, which, combined with depth distortion and normal consistency regularizers, markedly improves geometric detail and facilitates the extraction of noise-free meshes. While highly effective, purely planar 2D primitives can be insufficient for complex real-world objects that contain both flat surfaces and volumetric details, such as fur or foliage. A mixed-primitive framework that jointly optimizes both 2D disks and 3D ellipsoids has been found to address this, using a compositional splatting strategy and vertex pruning to adaptively select the appropriate primitive geometry for each region, leading to more accurate surface reconstruction on objects with hybrid structures [78].

Beyond modifying primitive shape, an alternative line of work integrates explicit or implicit geometric constraints to enforce the underlying surfaces. A distinct subcategory aims to anchor Gaussian splats directly to an underlying mesh structure. The hybrid mesh-Gaussian representation exploits textured, texture-rich flat regions to handle high-frequency color variations with significantly fewer parameters, while retaining Gaussians only for intricate geometric regions [79]. This architecture yields a final output that directly provides a textured mesh as byproduct, achieving higher FPS and reduced computational overhead. Despite these advances from mesh-anchored and planar methods, accurate surface extraction from a Gaussian cloud without explicit surface-aligned primitives remains viable. A promising strategy termed render-then-fuse bypasses direct reliance on the geometric positions of the Gaussian elements, instead capitalizing on the strong novel-view synthesis capabilities of the model. By rendering pairs of stereo-calibrated novel views and applying a stereo matching method, high-quality depth profiles are extracted and subsequently fused via TSDF integration into a watertight, geometrically consistent surface [77]. This insight that the intermediate 3DGS parameters may be redundant for surface extraction in favor of the rendering output provides a remarkably robust alternative to extraction from Gaussian positions, particularly for in-the-wild and large-scale scenes.

Yet, robust surface reconstruction ultimately depends more on geometric understanding than the Gaussian positions themselves. In that direction, a geometry-aware Gaussian method redefines the rendering process by computing depth via differentiable ray-ellipsoid intersection rather than from Gaussian centers [16]. This analytic depth formulation provides an explicit gradient pathway to optimize the rotation and scale attributes of Gaussians yielding high-quality surface normal estimation, making the optimization landscape clearer for downstream tasks like mesh and normal extraction. Ultimately, the field is converging on a hierarchy of sophistication: from modifying primitive kernels (e.g., planar or mixed primitives), to binding them to explicit anchors (meshes), to using the rendering pipeline itself for depth fusion. The trade-offs are profound—the mixed-primitive methods achieve speed, while the 2D planar and mesh-hybrid methods achieve higher extraction quality. The emerging trend, clearly successful, is towards decoupling geometry from appearance, allowing the Gaussian optimization to focus on photorealistic rendering while a geometrically constrained representation provides the structural stability. For the future, we envision hybrid frameworks that combine the segmentation ability of 2D Gaussians with learned surface refinement will push toward millimeter-accurate reconstruction, bridging the widening gap between photogrammetry and real-time interactive simulation.

4.2 Physically-Based Modeling and Material Decomposition

The evolution of 3D Gaussian Splatting from a purely photometric scene representation to a physically-based rendering (PBR) framework marks a critical advancement toward applications requiring relighting, material editing, and realistic light transport. While the preceding section established explicit, surface-aware geometry as the foundation for structural stability, this subsection addresses the complementary challenge of *appearance decomposition*: re-parameterizing Gaussian primitives—traditionally defined by position, covariance, opacity, and spherical harmonic (SH) coefficients—to instead encode intrinsic material properties and disentangle the scene into geometry, materials, and illumination. This decomposition is not merely an aesthetic enhancement; it is a necessary condition for interactive editing, relighting, and physically faithful simulation. The core challenge lies in the inherent coupling of appearance and geometry within the Gaussian primitive, where the view-dependent color must now be resolved through a physically-inspired rendering equation rather than a purely emissive model.

Initial attempts to model complex appearance replace the limited frequency capacity of SH with alternative basis functions. Spec-Gaussian, for instance, employs anisotropic spherical Gaussian (ASG) fields to capture high-frequency specular components [80], demonstrating significant improvements in rendering fidelity but stopping short of a fully disentangled material model. To achieve true intrinsic decomposition, differentiable PBR frameworks build upon the splatting pipeline by introducing per-primitive material maps and a lighting model. The core principle is to express the outgoing radiance as a function of geometry (normals, roughness), material (albedo, metalness), and an incident illumination field. For instance, Bidirectional Gaussian Primitives adopt a light- and view-dependent scattering function via bidirectional spherical harmonics, allowing the representation to unify surface and volumetric behavior under dynamic illumination [10].

A significant line of research focuses on improving the representation and disentanglement of surface materials to better model complex light phenomena. Methods can be broadly categorized by their treatment of material and light. One approach introduces **per-primitive texture maps** to model spatially varying materials. Drawing an analogy to classical texture mapping, GStex textures each 2D primitive to decouple appearance from geometry, allowing a single Gaussian to represent fine texture details and significantly reducing the primitives needed for smooth geometry [7]. Further research like ASAP-Textured Gaussians and Content-Aware Texturing optimize texture allocation and resolution, demonstrating that adaptive sampling and anisotropic parameterization can achieve high-fidelity rendering with fewer parameters by concentrating texture capacity on high-contribution regions [81, 82]. To this end, TextureSplat uses per-primitive textures for spatially varying normals and materials to model highly reflective surfaces and utilizes a GPU-accelerated material texture atlas for efficient rendering [9]. SVGS and Gaussian Billboards similarly employ spatially varying colors via texture maps or interpolation on 2D Gaussians to enhance expressiveness while maintaining strong geometry [83, 8]. Additional explorations with Textured-GS and Textured Gaussians also propose augmenting primitives with RGBA textures for richer appearance, demonstrating gains in visual fidelity [26, 25].

A second generation of methods integrates PBR with a more explicit lighting model. Normal-GS proposes the use of a physically-based rendering equation, where the color is re-parameterized as a product of a normal vector and an Integrated Directional Illumination Vector (IDIV), showing that incorporating normals into the pipeline improves the fundamental accuracy gap between geometry and appearance [84]. In parallel, the most notable advancement in material modeling has been the inclusion of an explicit BRDF decomposition, but such decomposition may restrict the representation of view-dependent appearance. Mesh-based, ray-traced methods such as MPG (Geometry Field Splatting with Gaussian Surfels) are effective for sharp specularities in Gaussian fields, an area where SH is a clear failure mode [27]. These methods integrate material attributes as per-primitive attributes, and often existing work emphasizes the importance of accurate surface geometry to make material disambiguation accurate. For instance, Neural Shell Texture Splatting disentangles geometry and appearance by sampling a neural texture field for high parameter efficiency and easy textured-mesh extraction [13]. The partition of the Gaussian field into learnable regions, as offered by Spherical Voronoi, provides a framework for appearance encoding in a PBR model. It proposes learnable reflection probes, taking reflected directions as input following principles from classical graphics [85].

A critical component of material decomposition in Gaussian Splatting is the estimation of surface normals. Gaussian renderers are frequently purely visual, and their appearance is not directly conditioned on normals. Work that integrates normals into the rendering path, such as Normal-GS [84], uses optimizable normals as part of the physically-based shading equation, which implicitly regularizes the material. The joint work on geometry field splatting uses splatted geometry and latent representations with SH encodings of reflection vectors rather than encoded colors, achieving significant improvement in surface reconstruction quality on widely-used datasets [27]. Additionally, Neural Shell Texture Splatting reduces the number of primitives but still requires coarse geometric priors to place textures; this is a common trade-off in texture-augmented GS approaches [13].

Finally, **an important formalization puts emphasis on the efficiency of material representation to improve usability in editing.** Decomposing a scene into a discrete finite set of shared material attributes such as textures and a parametric appearance model creates editable and compact representations. The assignment of spatially varying properties via opacity or RGB textures, as explored in Textured-GS and Spatially Varying Gaussian Splatting (SVGS), enhances scene editing and re-texturing tasks [26, 83]. In contrast to "image" level color transforms, [86] addresses appearance/lighting variations in a continuous manner, not fully interactive lighting effects [86].

The entire domain of physically-based radiance fields requires careful integration of geometry and optical radiance. Research is now moving towards the automatic conversion of radiance fields into stable and disentangled material properties. To extract a proper physical normal, surface estimation via geometry fields from Gaussian Splats—e.g., perspective-accurate 2D splatting—is paramount [6]. Moreover, existing works are often combined with optimization strategies and representation learning that harnesses the volumetric nature to parameterize appearance. From the observation that Gaussians are volumetric and (usually) unordered representation, PBR Gaussian splatting can be used as a robust, unstructured radiance field. However, the pursuit of material independence for all possible light, reflection, and semi-transparencies in physical environments highlights the limitations of the impulse function for high-frequency appearance and geometry [87, 50], while [49, 78, 66] show the efficacy of coarser approximations. The acceleration and simplification of plausible, relightable GS representations will open frontiers for AR/VR and digital twins, enabling workflows that were previously inaccessible from the neural point of view. The final objective for the community is to make it as simple to generate physically-based materials from real-world scenes as it is to capture radiance fields, unencumbered by geometric imprecision and non-disentangled appearance.

4.3 Spatial Scalability and Large-Scale Scene Representation

The transition from object-centric reconstruction to unbounded, city-scale environments fundamentally challenges the memory hierarchy and computational model of 3D Gaussian Splatting (3DGS). Unlike bounded scenes that fit within a single GPU's memory, unbounded environments contain millions of primitives whose combined parameters far exceed on-device capacity, while real-time interaction demands immediate access to only the visible subset. Spatial scalability thus hinges on two complementary strategies: (i) architectural designs that organize primitives hierarchically to enable selective access, and (ii) distributed or out-of-core pipelines that schedule computation and memory movement across devices. These are not mutually exclusive; rather, practical systems integrate both to balance training tractability, memory footprint, and rendering speed.

Early approaches to scaling 3DGS adopted a divide-and-conquer strategy for training. By partitioning the scene into blocks—using load-balanced KD-trees or spatial hashing—the optimization process can be distributed across multiple GPUs. This paradigm distributes primitives and associated training data across nodes, requiring synchronization protocols and communication-aware scheduling. The key challenge here is achieving load balance, typically addressed by partitioning based on camera counts per block or by visibility culling to eliminate redundant work between partitions. Such distributed training preserves the high quality of per-block optimization while enabling a linear scaling of representational capacity with the number of devices, though it introduces the overhead of network communication and the risk of seam artifacts at block boundaries.

Dissociating from the GPU's memory entirely, **out-of-core optimization** tackles the bottleneck of data transfer. Instead of distributing the scene across devices, these approaches treat the CPU RAM as a large, slow memory pool, offloading the Gaussian parameters to host memory and streaming only the data needed for the current optimization step. This perspective positions the trained constitutive as a paged repository, where a fine spatial granularity allows only relevant primitives to be fetched. Such systems directly address the maximum scene size tractable on a single workstation and decouple the optimization scale from GPU memory constraints. A key requirement is an efficient visibility query during training to predict which Gaussians will be rendered from the current cameras, enabling a minimal and rapid data transfer to the GPU. This approach avoids the complexity of distributed synchronization while achieving significant scale with a single machine.

Once a scene exceeds device memory, **out-of-core rendering** becomes critical and relies on spatial data structures—most commonly level-of-detail (LOD) hierarchies. These structures organize primitives at multiple spatial granularity, enabling the renderer to select a subset based on camera distance and target resolution, thereby bounding the number of active Gaussians per frame. Hierarchical, voxelized or octree-based structures store primitives at a coarse granularity, with a selection mechanism that filters and streams the necessary data to the GPU. The efficiency of such systems is often enhanced via temporal and spatial coherence and tile-based traversal, where pre-computed visibility and streaming predictions pre-fetch required data. [88] Complementary to this, acceleration of the rendering core through hierarchical spatial partitioning reduces the per-frame cost of dense regions and balances the warp workload. These scalable software approaches are well complemented by system-level improvements that accelerate the pipeline on constrained devices, such as offloading culling to an earlier preprocessing stage and prioritizing high-contribution primitives [89].

Looking ahead, the field must reconcile the increasing scale and fidelity of reconstructed scenes with the tight memory and bandwidth constraints of resource-constrained and real-time platforms. A promising direction is the development of *progressive* representations that support *streaming with continuously adaptive granularity.* The goal is to evolve from static, post-training LOD structures to adaptive systems that prioritize, fetch, and evict primitives online under a bandwidth budget. The stronger integration of scene partitioning with the renderer, where the rendering process itself drives the data schedule of training and streaming, will be central to realizing truly scalable 3DGS. This convergence of algorithm and system design is the necessary step to bring high-fidelity city-scale world models to interactive platforms.

4.4 Compression, Compact Representation, and Storage Optimization

The substantial memory footprint of trained 3D Gaussian Splatting (3DGS) models remains a critical bottleneck for deployment in bandwidth-constrained and real-time applications. While the previous subsection addressed how spatial scalability can be achieved through hierarchical organization and out-of-core streaming, a trained scene—even when efficiently partitioned—often comprises millions of primitives whose raw parameters, including positional data, covariance matrices, opacity values, and spherical harmonic coefficients, require hundreds of millions of floating-point values. This subsection surveys the rapidly evolving landscape of compression and compact representation techniques, focusing on two complementary levels of the pipeline: the refinement of density control mechanisms to reduce primitive count, and the bit-level reduction in parameter storage through quantization and entropy coding. Moreover, the importance of these techniques extends beyond standalone deployment: as noted in the following subsection, more compact intermediate representations directly reduce the input complexity for feed-forward generalizable models, making compression collaboratively beneficial across the broader 3DGS pipeline.

A foundational strategy for reducing model size is to improve the efficiency of the primitive set itself through more principled density control. The original heuristic Adaptive Density Control (ADC) often generates redundant or overfitted Gaussians. More compact and accurate models can be produced by replacing or augmenting positional gradient heuristics. For instance, methods like [53] advocate for a pixel-error-driven criterion for densification, enforcing a controlled primitive count and correcting bias in opacity handling. [57] offers a theoretical framework for densification, establishing necessary conditions for splitting and identifying optimal parameter update directions to minimize loss while maintaining a compact point set. Similarly, [90] proposes long-axis splitting and recovery-aware pruning to better enhance primitive properties. The [55] technique uses opacity gradients as a lightweight proxy for rendering error, achieving dramatic reductions in primitive count. [54] leverages the angular coherence of gradients to avoid redundant splits, while [91] establishes a direct relationship between Gaussian scale and frequency content, deleting primitives of improper scale. [56] further improves Gaussian allocation by merging optically-similar primitives. Beyond heuristic improvements, the field is also witnessing a paradigm shift toward fully learnable policies, as exemplified by [58]. This controller learns when to split or prune primitives by quantifying their marginal contribution to the reconstruction, striking a more sophisticated balance between quality and compactness. To standardize comparisons among such strategies, [59] introduces a fixed-primitive budget, harmonizing densification and pruning cycles across methods for capacity-matched evaluation—this is particularly relevant for downstream tasks where the primitive count directly influences the efficiency of subsequent streaming or out-of-core rendering.

Beyond reducing the number of primitives, significant efforts have focused on the bit-representation of the remaining parameter set, using vector quantization (VQ), entropy coding, and pruning. These methods often frame compression as a rate-distortion optimization problem, allowing principled trade-offs between model fidelity and storage. While hierarchical spatial structures from the previous subsection facilitate selective access at runtime, quantization and entropy coding ensure that the primitives—once selected—occupy minimal bits, thus complementing the paging-level scale reduction without compromising the interactive access. Conversely, it is worth noting that compression itself aligns naturally with the scalable rendering discussed earlier: a smaller per-primitive footprint effectively increases the number of primitives that can be streamed or fit within an LOD-based working set, thereby improving cache hit rates and reducing bandwidth pressure. Other efforts explore the most aggressive levels of pruning, distilling the entire scene representation into a neural shell or low-dimensional embedding, which intimately couples geometric fidelity with appearance. The specific choice of machine learning architecture and compression algorithm is often application-specific, balancing storage size, rendering speed, and peak signal-to-noise ratio.

The confluence of principled density control, quantization, and entropy encoding forms the basis of a modern, scalable 3DGS pipeline. Within the broader context of this survey, compression is not an isolated post-processing step but a core necessity for enabling both out-of-core exploration and the practical deployment of learned, forward-predicted scenes. Looking forward, the next frontier lies in a holistic approach, where density control, attribute quantization, and entropy encoding are jointly optimized end-to-end, and where a tight interaction with the training process allows the model to converge to stable local minima tailored to the chosen compression strategy. Moreover, for the *initialize-and-refine* paradigm described in the following subsection, compression-aware system design will become indispensable, as the size of learned priors—when partially trained or distilled—directly impacts their reuse as fast-adaptable initializations. Ultimately, the goal of such cross-cutting methods is to enable the generation of digital 3D content with maximum fidelity per bit, ensuring it can be delivered efficiently and interactively to a wide range of devices, from the edge to the cloud.

4.5 Feed-Forward and Generalizable Reconstruction

The transition from per-scene optimization to data-driven, generalizable reconstruction represents a fundamental paradigm shift in 3D Gaussian Splatting (3DGS), addressing the critical bottleneck of requiring either reliable Structure-from-Motion (SfM) initialization or lengthy per-scene training. This subsection examines the architectural innovations and training strategies that enable learning-based systems to predict Gaussian scene representations in a single forward pass, thereby unlocking real-time reconstruction and cross-scene generalization.

A central research thrust focuses on feed-forward Gaussian attribute prediction, where deep networks directly regress the primitive set and its associated attributes—position, opacity, covariance, and spherical harmonic coefficients—from posed input images. An early consideration in this direction is the dependency on high-quality point cloud initialization; methods such as [92] propose structural attention mechanisms to achieve direct 3D reconstruction from random or noisy initialization, bypassing the constraints of accurate SfM. To further mitigate the "garbage-in, garbage-out" problem and produce dense, high-quality starting points, [32] triangulates pixels from dense image correspondences in a one-step geometry approximation, and [61] learns data-driven priors from large-scale indoor scenes to predict dense 2D Gaussian initializations, demonstrating that data-driven priors can provide the strong priors necessary to accelerate convergence and improve geometric fidelity in a way that SfM-based initialization often cannot. These approaches illustrate a broader shift away from processing-driven initialization and toward purpose-built, learned initialization networks and foundation models, as demonstrably shown by [63], which bypasses SfM entirely for an initialization procedure enabling fast convergence in as few as 50-60 views.

Building on the concept of a stronger initialization, a substantial body of work investigates how to design the prediction module itself for cross-scene generalization in both feed-forward and sequential settings. [93] offers a systematic study that reveals a dense initialization with strong densification does not guarantee visual improvements; instead, point-cloud density and structural soundness become decisive when aiming for view generalization and geometric consistency, adding a critical, quantitative layer to the current speculation about where the true value of learned, dense initialization lies. Similar insights are leveraged by [94], which positions well-designed initialization as a *decisive factor* for sparse-view 3DGS; their work augments a sparse SfM point cloud through frequency-aware augmentation, 3DGS self-initialization, and point-cloud regularization to compensate for regions that SfM fails to cover, positing that a better point cloud—not just better constraints—is the main route to generalizable performance.

Conversely, other works question the necessity of SfM quality altogether, demonstrating that carefully designed random initialization, combined with a robust optimization scheme, can often match or surpass the quality of a resulting SfM-initialized model [95]. This perspective is supported by robustness-focused pipelines that view initialization as a denoising process, ensuring coherent, stable spatial structure robust to noisy, sparse, or imperfect settings [96]. In the observer scenario, [97] demonstrates a Propose-Refine prototype that merges frames as they arrive with overlapping images via Local & Semi-Global optimization, showing partially initialized systems can be pushed to near-interactive rates without a global SfM.

A common consensus across many works is that a feed-forward prediction stage does not *perform* per-scene optimization, but may drastically expedite it. A two-stage schema is emerging as seen in [98], which: without SfM, frames coarsely posed images and noisy Lidar point clouds into a constrained decomposition that sequentially solves for the device-center-to-world and camera-to-device-center components, jointly estimating both the reconstruction and the parameters in a bi-level schema. Similarly, approaches that couple a learned, data-driven prior with transient-aware controls, as in [38], can separate static and dynamic modes in a robust manner from untrustworthy initializations. This interplay of robustness and visual realism is reinforced by the structure of [60], which provides a globally organized, region-prioritized framework to ensure structured spatial distribution and fast convergence; this being a critical property for downstream predict-then-refine pipelines where it is possible to reduce the initial variance of the predicted primitives and improve data flow in an optimization-aware way, as also highlighted in [64] which identifies local suboptimal solutions as a fundamental limitation on the path to high-quality reconstruction.

An emerging design principle binds all these works: *initialization as a trainable module*. The field is moving from ad-hoc point clouds (SfM) toward purpose-built neural predictors—triangulated dense correspondences [32], learned dense 2D Gaussian initializations [61], and foundation-model-derived priors for pose and geometry [63]. That this is not just a design choice, but a strategic necessity, is validated by studies showing that geometric consistency and generalization to off-trajectory views are heavily affected by the quality and distribution of the initial representation [93]. Further refinement is achieved through degradation-aware learning or constrained photometric-geometric optimization [96, 98], which makes it possible to bridge the gap between learned statistics and scene-specific optimality.

In conclusion, the state of feed-forward and generalizable reconstruction is trending toward "initialize-and-refine," where the *initialization is learned*, not given. The optimal design of such learned priors—as well as the interplay between the initial spread and the use heuristic-free, learned densification—remains one of the most fertile and open research avenues. An important, parallel line of work explores how better initialization can relax the convergence constraint and help avoid local optima [99, 64, 100]. The field's ultimate goal is thus likely a unified optimization paradigm that can trigger instant and stable convergence in dynamic, sparse, and extensive 3D scenes—categories, including where and how we train these learned priors, that will define the broad field of view synthesis and 3D content creation.

5 Dynamic Scene Modeling and Spatial-Temporal Representation

5.1 Foundations of Deformable and Canonical-Space Modeling

The dominant paradigm for dynamic scene reconstruction with 3D Gaussian Splatting (3DGS) is the canonical-space deformation framework. Rather than optimizing a unique set of primitives for every timestep, these methods maintain a single, static collection of 3D Gaussians in a canonical configuration and learn a time-conditioned deformation field that maps this canonical state to the observed scene at any requested moment. This architectural design amortizes the temporal modeling cost, ensuring that the system is parameter-efficient and inherently reuses geometric and appearance information across all frames [101, 102]. By shifting the burden of dynamic motion modeling to a potentially compact neural network, these frameworks represent a critical departure from direct per-frame optimization and serve as the foundation for many state-of-the-art dynamic reconstruction systems [1].

The deformation network takes as input a spatial coordinate concatenated with a temporal query (e.g., a frame index) and is trained to output the corresponding six-degree-of-freedom (6-DoF) displacement for a Gaussian primitive. This maps the canonical primitive's position and rotation to its target state using a rigid transformation, complemented by a residual update to refine the primitive's attributes. The functionality can be formalized as below:

"' θ_t = f_θ ({μ, t, γ(t)}) = {$\Delta\mu$, Δr, Δs} "'

where θ represents the deformation attributes, γ(t) is a temporal positional encoding, and $\Delta\mu$, Δr, and Δs denote the predicted position, rotation, and scale alterations. The deformed 3D Gaussian is then expressed through `μ_t = μ_can + Δμ` and `Σ_t = R_t S_t S_t^T R_t^T`, with `R_t` and `S_t` being the rotation matrix and scaling matrix derived from the deformed attributes. Concurrently, the opacity and spherical harmonic (SH) coefficients can be either pronounced per frame or implicitly predicted by the network, allowing for the reconstruction of time-evolving appearance effects [102]. This core formulation, of bypassing the direct optimization of a temporal attribute field to a function of a canonical set, has led to a broad architectural evolution.

Efforts to further optimize and scale the deformation module itself have produced rich models. The foundational approaches in this domain make use of plain multi-layer perceptrons (MLPs) with positional temporal encoding as timestamps, which serve as the baseline for motion approximation, staying novelty-proof and memory-efficient [101, 102]. However, capturing long sequences or highly complex dynamics with moderately simple architecture can lead to efficiency bottlenecks and may cause limited generalizability. To address this, subsequent literature has pivoted towards hybrid representations that separate spatial abstraction from time-variant features. Systems with deformation fields embed spatial and temporal information into a compact feature space, generating the final deformation vectors [19]. The explicit geometric prior of these grids allows for a more manageable scalar field and improves performance over pure MLPs, but introduces a non-negligible overhead through voxel interpolation and volume update, highlighting a common critical trade-off between representation capacity and rendering frame rate [103]. A notable advanced design incorporates a more recent, continuous-time deformation field, expressing the offset as the solution of a neural ODE or a variation with Closed-form Continuous-time (CfC) cells. This formulation enforces intrinsic temporal smoothness through the model by construction, rather than relying on post-hoc regularization. Such continuous-time dynamics enable stable interpolation between frames and even extrapolation into the unseen temporal domain, effectively avoiding arbitrary overfitted artifacts [101, 2].

Crucially, the learning dynamics of these fields are capitalized on the initial point propagation. The most common and robust initialization of the canonical 3D Gaussians is the sparse point cloud generated by Structure-from-Motion (SfM) at a fixed decoded frame [17]. This prior grounds the elapsed optimization so that the scene geometry are following depth cues and the canonical basis is well-constrained. Yet, while the novel view synthesis results in dynamic scenes show the prime strengths, the canonical-space architecture reproduces and even heightens a number of fundamental issues—stable optimization, local minima, and motion regularization—that find prominent solutions within the deformation-based methods [101]. Sparse or poor-quality SfM point initialization often results in dark, degenerate deformation fields in under-observed regions, and unsupervised tuning of the point budget tend to create uncorrectable clusters. Likewise, gradient-constrained optimization on the dense set of primitives leads to the classic "floaters" effect or blur traps, where background dynamics are misconstrued. To mitigate these, subsequent works have integrated depth priors—monocular or multi-view—and initialised new primitives in texture-less areas [104, 101].

A more recent set of techniques has brought even greater refinement to guaranteeing the topology in this mapping framework. Spatial rigidness constraints are imposed to preserve the local neighborhood of a Gaussian group across time, ensuring that the primitive is moved under a consistent local rigid transformation—crucially preventing unrealistic geometric changes in regions where the temporal magnitude is high [19, 105]. Another common source of learning bias is motion ambiguity: single view and absence of severe occlusion produce indeterminate temporal flows; to resolve these, works seek additional, implicit and explicit, motion priors—incorporating temporal consistency or depth and optical flow estimates into a geometric error weight [101]. Finally, implementable and expeditious engineering to ease optimization in weakly super vised regions indeed relies on a confidence-based training approach to calibrate the deformation with noisy memory, proving that the final velocity field and quality of the scene are sensitive [103].

In summary, the canonical-deformation scene remains a testament to the broader theme of explicit/implicit modeling in 3DGS. The explicit point set handles the essential parallax and appearance fidelity, whereas the deformation network acts as a controller of the motion field—which continues to tune the trade-offs between speed, quality, storage and temporal robustness [1, 1]. As these methods evolve to borrow more scalability from content-aware models, distributed frames, and the long-horizon (i.e., continues over thousands of timestamps), an open challenge remains to maintain consistent canonical propagation, and even more, a deep and compact canvas where a hierarchy of dynamic data can uniformly flow. Future works will likely solve multi-frame warping and rely on continued dynamism to extend this amortized framework to what could become arbitrary long, possibly sequence-coherent, dynamic environments.

5.2 Explicit and Native Spatio-Temporal Gaussian Representations

A significant paradigm shift in dynamic scene modeling departs from the canonical-space-plus-deformation approach by treating the 4D spatio-temporal volume as the fundamental domain for primitive definition. While the canonical deformation framework of the previous subsection amortizes temporal modeling through a separate network, these methods directly or natively embed time into the Gaussian parameterization, aiming to bypass the architectural overhead of deformation MLPs and achieve inherent temporal coherence through explicit geometric constructs. The motivation for this shift is twofold: first, to decouple the optimization of static appearance from potentially complex motion mappings, and second, to enable a unified representation that naturally handles both continuous, real-time rendering and temporally dense reconstructions. In this sense, the paradigm serves as a direct architectural alternative—one that trades the flexibility of learned deformation for the compactness and interpretability of time-native primitives, which in turn informs how the following subsection decomposes scenes into static and dynamic parts for efficiency.

Within this paradigm, a primary line of work introduces true 4D Gaussian primitives, extending the mathematical formulation from spatial to spatio-temporal covariance. Instead of displacing a static point, the primitive's center and covariance are defined as a 4D structure in a joint space of spatial coordinates and time. For rendering, these are projected into the volume at a given timestamp by slicing the 4D kernel, effectively predicting the 3D shape that emerges from the intersection of the hyperplane [11]. To model time-varying, view-dependent appearance, these methods introduce specialized basis functions, such as Spherindrical harmonics, a generalization of spherical harmonics defined over Cartesian and polar coordinates within the spatial and temporal domains, allowing the anisotropic radiance to be a joint function of both view and time. The direct formulation of a 4D primitive can present significant computational overhead, as the covariance matrix becomes 4D, requiring higher memory for storage and projection calculations. Yet, this formulation is appealing because it serves as a direct contrast to the canonical-deformation approach by avoiding the explicit separation between geometry and motion, offering a compact and theoretically cleaner alternative.

Recognizing the complexity of full 4D primitives, another line of research implements time-augmentation to existing 3D Gaussians. Rather than treating time as a spatial dimension, these methods augment the 3D primitives with explicit, time-dependent parameters, effectively allowing them to exist at multiple timestamps within a single framework. This cuboid representation allows for the existence of the Gaussian at multiple timesteps, parameterized by a cubic Bézier curve drawn to control the extent of temporal presence. This parametric avenue constructs a continuous spatio-temporal solid that defines the temporal trajectory of the Gaussians. By parameterizing the opacity, color, and position as functions of time, time-augmented methods capture dynamic elements while maintaining a clear, interpretable connection to the initial 3D representation. This stands as a less radical departure from canonical methods, yet still avoids the MLP bottleneck by placing the time dependence directly onto the primitive attributes—making it an intermediate step between the deformation-based and fully 4D-native frameworks.

Another set of techniques avoids the rigid structure of time-augmented frameworks and instead explores more flexible, non-symmetric, and time-aware kernels to represent the spatio-temporal volume. This category can be particularly effective for handling irregular motion not representable by symmetrically deformed Gaussians. The linear expansion of Skew-Normal distributions that introduce a bounded learnable skewness parameter to support half-Gaussian-like shapes is not restricted but generalized to produce anisotropic geometry, allowing the representation of the movement of an edge as it travels from one instant in time to another.

This extension yields physics-inspired anisotropic geometry to model temporal movement and is capable of reconstructing boundaries as they appear and disappear over time, as well as asymmetric scene discontinuities more closely than a symmetric Gaussian kernel. The key insight of this approach is that addressing non-linear motion requires inherent adjustments in the representation capacity to accommodate irregular temporal features, rather than merely adding a learnable time vector to a frozen 3D point cloud as done in prior paradigms.

Finally, the last category relies on dividing the temporal signal into a set of orthogonal frequency bases rather than treating space-time as a direct continuous dimension. This permits a logical division that controls motion through decomposition into a linear combination of temporal bases, pairing a separation into the low-frequency (rigid, large-scale motion) and the high-frequency (non-rigid, fine geometric deformation) components. This enables principled historical reconstruction geometric and interpolation over long sequences without the cost of full temporal sampling. For example, B-Spline bases with linearly expressed trajectories generate a set of linked, smooth control points, offering not only renderability but also implicit temporal resampling for novel view and time synthesis. In comparison to per-frame deformation in the canonical-space framework, such basis decompositions impose smoothness across the time axis and provide a compact, interpretable, and efficient representation of the dynamic scene while also providing robustness against temporal over-fitting. Yet, the choice of a basis set requires an *a priori* hypothesis on the frequency content of the scene's most appropriate dynamic features and may fail to express abrupt or non-periodic behavioral changes—highlighting a trade-off between representational efficiency and expressiveness, neither of which the static-dynamic decomposition of the following subsection can circumvent.

5.3 Static-Dynamic Decomposition and Multi-Object Scene Partitioning

The static-dynamic decomposition paradigm emerges from a fundamental inefficiency in monolithic 4D Gaussian representations: applying a uniform deformation field or spatio-temporal model to all primitives wastes substantial representational capacity on regions that remain geometrically invariant over time. This partition is not merely a computational convenience; it is a representational strategy that fundamentally alters the optimization landscape. By isolating persistent static elements from moving foreground objects, methods can deploy their expressive power where motion actually occurs, yielding improved optimization stability, reduced memory footprints, and enhanced scene understanding.

The most common approach for classification is the use of semantic priors derived from foundation models. Optical flow, semantic masks, and features from models like CLIP or DINOv2 are projected into 3D space and assigned to each Gaussian primitive to determine its dynamic status. This assignment, however, is not a mere binarization process—it carries substantial architectural implications. Methods assigning primitives to static and dynamic sets typically employ distinct branches: a compact representation for the static background, and a richer, deformation-aware branch for dynamic elements. From a representational standpoint, the static-dynamic partition mainly reduces memory: static regions—often significant portions of real-world scenes—can be stored with far fewer attributes, while dynamic foreground objects require additional parameters for covariance evolution, opacity tracking, and deformation fields.

Crucially, density control must also be partitioned. In dynamic scenes, a uniform densification heuristic that ignores temporal information often over-concentrates Gaussians in dynamic, high-motion regions, but then sparsely allocates poor detail to persistently occluded background areas. Addressing this, adaptive control techniques use temporal context to decide where to densify the primitives [88]. They focus on dynamic regions where the time-varying signal experiences fast and complex photometric changes, but may also revert to de-densification in static areas to control model complexity and avoid overfitting. However, such densification strategies require careful confidence and reliability estimation; noisy or unstable points from transient occluders or fast-moving objects can corrupt the optimization. Thus dynamic-aware heuristics—integrating visibility, reliability, and geometric confidence—are essential, particularly in Simultaneous Localization and Mapping (SLAM) contexts.

The analysis transitions naturally from partitioning for efficiency to object-centric tracking and multi-object association, where the problem's objective extends to persistent identity and trajectory recovery. The challenge becomes one of co-segmenting a dynamic scene into its broad object classes and re-identifying the resulting primitives over time. End-to-end strategies adopt a holistic approach: reconstruct and track the 6-DoF motion of *all* dense scene elements simultaneously, while maintaining scene coherence. Given that dynamic objects may be partially occluded, appearance alone is insufficient for tracking. Thus reliability in dynamic tracking requires deriving pseudo-ground-truth data from trained deformable and static geometry that can be applied and related with appearance, complexity, and motion priors.

A further key trend in the decomposition is the integration of *physics-informed persistence* to manage these challenging occlusion scenarios. By leveraging differentiable rigid-body non-rigid (e.g., position-based dynamics) simulations or projective assignment, the method can use physical principles to project object trajectories during long windows of full occlusion, and rely on collision heuristics to track and assign information to the occluded primitives during interactions. Even highly dynamic scenes contain strong physical priors regarding object persistence, and incorporating rigid behavior constraints minimizes the spatial extent of modeled Gaussians while ensuring temporal continuity.

Looking forward, a meaningful challenge is lowering the reliance of current decomposition methods on heavy annotation signals or specialized tracking objectives. Research is moving toward unsupervised estimation of clusters and normals from the expectation of temporal gradients and scene coherence of appearance streams. In these emerging systems, the static-dynamic partition can be leveraged as a principled 3D Gaussian-based representation for downstream tasks such as re-lighting, inverse rendering, or object manipulation. Future decomposition frameworks may be built toward more generalizable and less capture-conditioned generation of the 4D dynamic world.

5.4 Semantic, Controllable, and Interactive Dynamic Modeling

The integration of semantic understanding and controllability into dynamic 3D Gaussian Splatting (4DGS) represents a critical frontier, transforming these representations from passive photometric reconstructions into editable, interactive, and semantically aware digital assets. Building upon the static-dynamic decomposition established in the previous subsection—which separates persistent scene structure from moving elements—this subsection examines methodologies that further endow dynamic Gaussian scenes with high-level context, hierarchical motion decomposition, and explicit control signals, enabling intuitive manipulation, re-animation, and semantic querying. The core challenge is no longer merely capturing what a scene looks like, but also understanding what it contains, how its components relate, and how they can be intentionally altered. Crucially, this semantic layer is not additively superimposed; rather, it draws its robust foundations from the spatially and temporally complete reconstructions that uncertainty-aware regularization and dense initialization—introduced in the preceding subsection—ensure, making the semantic reasoning reliable even under casually captured, sparsely initialized conditions.

A foundational leap in this direction involves the infusion of semantic features from vision-language foundation models into the spatio-temporal Gaussian structure, a process often termed Semantic 4D Gaussian Splatting. These models embed per-primitive representations that align with the feature space of models like CLIP, enabling open-vocabulary queries and semantic segmentation directly within a dynamic volume. A key advancement in this domain includes hierarchical flow decomposition and dual-hierarchical optimization to disentangle the guidance, which separates flow signals from semantic cues for dynamic foregrounds, enabling more robust understanding. This semantic grounding is not merely a descriptive tool; it forms the basis for part-level decoupling and association across the temporal domain. In a manner conceptually analogous to the static-dynamic partition (but operating at a finer granularity, within the dynamic set), it converts the raw Gaussian cloud into a structured, hierarchical one that can be reasoned about, taking the first step from "what is moving" to "what is this object and what are its constituent parts".

Building upon the ability to separate and semantically label scene components, part-aware deformation and motion-guided manipulation become possible. Methods in this area move beyond global deformation fields to decompose dynamic scenes into part-level groups based on motion cues and differentiable segmentation. This decomposition enables several forms of user-directed control: an object part can be represented with an identity encoding vector that controls the trajectory of the object's centroid, enabling direct trajectory editing, or users can apply specified transformations to specific parts to drive re-animation. However, integrating these affordances requires navigating inherent trade-offs. Direct per-part rigid-body constraints can minimize the computational cost of computing transformations but may restrict the expression of nuanced, flexible deformations at region boundaries. Conversely, dense, flexible deformation offers greater flexibility but adds the risk of motion ambiguities, where distinct motions are mapped to the same pose, harming temporal stability.

In an interactive and controllable context, hierarchical motion decomposition becomes the pivotal intermediary between human intent and low-level optimization. The ability to disentangle the motion of an entire articulated object from the intricate movement of its individual parts is what allows a user to either often reposition a character or fine tune the wave of a hand. This capability is, nevertheless, contingent upon an initial optimization that is sufficiently robust to yield a clear, motion-consistent decomposition; without it, subsequent editing will naturally propagate errors. In this sense, the robustness mechanisms of the previous subsection—uncertainty-aware regularization and density augmentation—serve not only reconstruction quality but the integrity of the edits, since cleanly rendered dynamics are what produce interpretable semantic cues from which parts can be extracted. The control structures described above also enable direct user interaction, where a user can apply a transformation to a part at a single point in time and observe the effect distributed seamlessly through the sequence.

Looking towards the future, the field is set on a path towards longer-temporal coherent modeling and more open, user-specified semantic control. Current challenges can be traced back to the memory cost of these hierarchical representations, especially for long video sequences, a burden that is exacerbated when dense semantic descriptors are stored per–primitive. The most promising pathways involve strengthening ties with foundation models for zero-shot control and larger-scale semantic discovery of scene components, relieving the need for manually-annotated part labels. Furthermore, the advent of continuous-time representations suggests a future where arbitrary "retiming" is possible (e.g., producing slow-motion effects) via user-manipulation of a continuous trajectory. Thus, the future of 3DGS is one where semantic discovery, physics-informed priors from the static/dynamic domain, and the robust uncertainty-aware gains from uncasingly-casual inputs converge into a representation—built from the ground-up based on observation noise and uncertainty, yet delivering semantic stability and control that goes beyond the seeing to a understanding of the dynamic world.

5.5 Occlusion, Robustness, and Monocular Video Degradation

The reconstruction of dynamic scenes from casually captured monocular videos introduces a distinct set of challenges that are largely absent in controlled, multi-view or static settings. These challenges stem primarily from the fundamental ill-posing of the problem: temporal motion induces uncertainty, high-frequency dynamics often outpace the frame rate causing aliasing, and the reliance on Structure-from-Motion (SfM) frequently leaves dynamic regions devoid of initial geometric anchors [95, 93, 94]. Consequently, optimization pipelines must be robustified against unobserved regions and high-frequency time variations to prevent severe artifacts such as ghosting, floaters, and blurry temporal renderings. A primary avenue of research addresses these issues through uncertainty-aware regularization, which quantifies where the optimization can be trusted. Given that dynamic regions are exactly those areas that lack reliable SfM initialization and consistent photometric cues, these methods are needed to apply strong geometric priors selectively based on confidence. For instance, by leveraging pre-trained depth priors, but only activating them in areas of high spatial-temporal uncertainty, these techniques prevent overfitting to noisy photometric signals and mitigate the ghosting artifacts that arise from partial observations [98, 38]. This approach explicitly minimizes the adverse impact of under-constrained and ambiguous regions in the reconstruction, which is particularly crucial for modeling non-rigid deformation.

To address the issue of SfM-empty dynamic regions, a second category of solutions emphasizes the use of denser priors for initialization and optimization, such as dense point initialization and flow-guided primitives [96, 32]. Instead of relying solely on a sparse feature-matching point cloud, which is known to be brittle for highly dynamic content, these methods leverage estimated dense depth maps and scene flow to densify the base primitive set. By seeding Gaussians on dense depth maps, the reconstruction has a foundational structure to begin the optimization process from. Similarly, for fast-moving regions, the optimization uses optical flow or a 3D scene flow field to guide the movement and densification of primitives. By enforcing temporal constraints via flow, the optimization does not need to rely on noisy initialization. For this reason, this type of guidance can successfully recover geometry for highly dynamic parts of scenes where SfM provides no initial anchors, leading to a more complete and coherent dynamic representation.

The robustness of dynamic modeling is further tested by severe degradation such as motion blur, which frequently corrupts real-world monocular sequences. This temporal high-frequency information is challenging for a model that uses a discrete temporal sample at the frame timestamp because the actual observed image intensity is the integral of the entire exposure time, mixing a variety of 3D shapes and appearances. To counter this, models of continuous motion were designed to predict the location of a primitive at a series of sub-frame times and integrate its color contribution. However, these methods provide physically principled mechanisms for modeling temporal blur and directly contributing to the formation of a sharp image representation—the corresponding approaches were not included in the references above, so such a discussion is omitted from the survey.

Finally, confidence-based training for fallback mechanisms addresses robustness in the higher-level pose estimation within the 3DGS optimization [63]. In degenerate cases such as large portions of a scene being under-observed or dimly lit, the optimization process can be brittle. This line of work similarly improves the resilience of the pipeline by introducing a feedback loop: a trained model analyzes both its own prediction error and different pose estimates, with the model expected to fall back on a more robust, but perhaps a less, accurate pose prediction method or local optima when there is high uncertainty. This dynamic selection of a pose prior during training increases the robustness of the reconstruction pipeline, allowing for convergence in domains that would likely get stuck in an incorrect local optimum if a single assumption is chosen. Through these combinations of uncertainty-driven regularization, better point density, and physically grounded degradation modeling, dynamic 3DGS frameworks are reaching a higher level of robustness and are capable of handling increasingly more complex and casually captured inputs. Future directions should consider a tighter integration of these layers, an example of coupling uncertainty to the pulse of the blur kernel or flow guidance as a single, closure-rich optimization loop [38].

5.6 Dynamic Surface Reconstruction and Simulation

Building upon the robust and uncertainty-aware dynamic reconstructions detailed previously, the next frontier is elevating these spatio-temporal representations from mere visual proxies to physically grounded, editable, and simulatable assets. This transition necessitates a critical convergence of photorealistic rendering with geometric fidelity and physical plausibility. While deformation-field-based methods [106, 107] excel at novel view synthesis, they often neglect the underlying surface structure necessary for downstream applications such as editing, animation, and physics-based simulation. This subsection examines methodologies that bridge this gap by coupling dynamic Gaussian scenes with explicit surface representations, enabling the creation of deformable, editable, and simulatable 3D assets that extend beyond the robustness gained from uncertainty-driven regularization and denser initialization strategies discussed above.

A foundational strategy extends the planar 2D Gaussian formulation to the temporal domain—a natural progression from the flow-guided densification that anchors primitives to scene geometry. By learning canonical 2D splats and deforming them through a time-conditioned mapping, these space-time approaches leverage depth and normal regularizers to enforce surface alignment, achieving high-fidelity dynamic surface recovery even under occlusion and complex topological changes. The key insight is that the planar primitive's inherent orientation provides a stronger geometric prior than volumetric ellipsoids, yielding more stable normal estimation and enabling coherent mesh extraction. However, the reliance on per-scene optimization and the potential for the deformation field to introduce tangential sliding remain open challenges, echoing the brittleness issues that uncertainty-aware regularization sought to mitigate at the earlier feature level.

To achieve more controllable deformation beyond what learned mappings can offer, researchers have introduced cage-based and control-structure methodologies. These systems bind Gaussian primitives to a coarse, user-defined control mesh or cage; transformations applied to the cage are propagated to the Gaussian cloud, with covariance matrices updated accordingly to preserve local anisotropic shape [108, 65]. This explicit control mechanism supports intuitive kinematic manipulation and large-scale free-form deformations, providing a clear advantage over pure learned deformation fields for interactive tasks. The computational cost lies in the initial embedding and ensuring the upsampled Gaussian kinematics remain consistent with the underlying surface, typically most efficiently managed through geometric constraints that reinforce the flow-guided constraints discussed earlier.

A more recent frontier integrates physics-based simulation directly into the dynamic Gaussian framework, addressing the physical plausibility that simpler deformation models often ignore. This is achieved via data-driven learned physics modules or structured constraints that provide residual dynamics to a simulated mesh surface. The strategy often involves a coupling interface: a set of "anchor" Gaussians are bound to a deformable mesh, which is then simulated using continuum-mechanics-based methods such as the Material Point Method (MPM) or Finite Element Method (FEM). The explicit nature of the Gaussian representation is leveraged to provide photorealistic visual updates within the prediction-correction loop, refining the physical state based on visual feedback. This co-design ensures visual and physical fidelity jointly, enabling the simulation of complex phenomena such as local non-rigid deformations and dynamic interactions—staying within the realm of robustness methods that adapt to high-frequency dynamics and inconsistencies.

Despite progress, several limitations prevail, echoing the SfM-free and sparse-view challenges of earlier subsections. The requirement for accurate initialization often relies on dense and consistent inputs, which sparse-view or monocular dynamic capture may not provide. A promising direction is the pursuit of generalizable frameworks that can deduce dynamic surface parameters from a few views in a feed-forward manner, bypassing costly per-scene optimization for real-time human avatars and interactive media. Furthermore, robustly addressing topology changes (e.g., tearing or merging) where a watertight mesh is a poor proxy remains an open problem [109, 38]. Future work will likely focus on hybrid representations that dynamically combine Gaussian primitives for appearance with implicit or explicit surfaces for geometry and physics, guided by adaptive density control and the robustness criteria established above. Thus, this progression from robust, uncertainty-driven representation to geometrically and physically anchored surface understanding forms the backbone for the original, simulatable, and generalizable dynamic scene generation described in the following subsection.

6 Scene Understanding, Interaction, and Editing Applications

6.1 Semantic Decomposition and Open-Vocabulary Scene Understanding

The integration of high-level semantic understanding into 3D Gaussian Splatting (3DGS) represents a paradigm shift from photometric fidelity toward scene comprehension, transforming the explicit primitive-based representation into a unified carrier of geometry, appearance, and meaning. This subsection examines the distillation of foundation model features—such as CLIP, SAM, and DINOv2—into Gaussian primitives, the construction of spatially coherent feature fields, and the mechanisms enabling natural language-driven querying, segmentation, and classification.

The foundational strategy for semantic 3DGS involves distilling pixel-aligned vision-language features into per-Gaussian attributes [50]. Rather than treating color and semantics as independent quantities, these methods augment each primitive with a high-dimensional feature vector that encodes semantic identity and open-vocabulary alignment with text embeddings. This design inherently exploits the explicit, point-based structure of 3DGS: discrete primitives provide both a sampling grid and a topological anchor for back-projecting 2D foundation model outputs into 3D space. The critical insight here is one of feature compression—while individual Gaussian primitives offer sufficient local capacity for color approximation via spherical harmonics, semantic features are inherently broader and more abstract, requiring careful distillation strategies to reconcile the granularity mismatch between 2D pixel-wise supervision and 3D primitive cardinality. An important characteristic of this embedding scheme is its ability to jointly optimize for photometric fidelity and semantic consistency, enabling a single forward pass through the rasterizer to produce both an image and a semantic map.

However, the naive per-Gaussian feature embedding suffers from a persistent challenge: cross-view feature inconsistency. When features are distilled independently from 2D foundation models, output at a specific view does not guarantee a globally coherent representation for a single object; the same surfaces observed from different viewpoints may acquire divergent feature embeddings. This fragmentation disrupts both instance-level segmentation and open-vocabulary queries, which rely on stable, view-invariant feature anchors. To address this, several research directions have converged on the development of unified, codebook-adopted feature fields and mask-centric contrastive learning objectives. Instead of optimizing features independently for each Gaussian, these methods couple feature learning with a shared codebook or latent space, promoting intra-object cohesion and inter-object distinction through contrastive pressure. In addition, 3D-conditional feature aggregation, performed on the fly using geometric proximity or established voxelized anchor scaffolds, establishes an implicit hierarchical feature refinement process that iteratively denoises the distilled signal using the underlying geometry. This geometric conditioning also addresses another fundamental issue: the cross-granularity mismatch between the localization capacity of a single Gaussian and the spatial extent of a semantic object. Semantic objects are compositional, spanning multiple dozen or hundreds of Gaussians, yet individual primitives do not have access to a holistic receptive field. By introducing local graph construction or structure-aware semantic backbones, the model can propagate locally redundant context across primitive groups without requiring a pre-existing segmentation mask.

We further note a distinct departure from conventional "semantic segmentation" towards a communication-constrained formulation of scene parsing. Gaussian primitives provide a communication graph analogous to sparse, unoriented point clouds. The question becomes: given a text description, can we compute spatial correspondences efficiently? The answer has evolved from simple nearest-text-embedding search to sophisticated, query-specific and context-aware decoding. For instance, open-vocabulary queries are handled by semantic feature fields that generate a distribution over class-agnostic text embeddings—allowing "red mug" and "ceramic object" to have orthogonal, yet overlapping, activation patterns in 3D. This nuanced behavior is an emergent property of visual-language models that is now faithfully inherited and adapted by the 3DGS representation. The explicit, discrete nature of 3DGS also introduces a novel research opportunity that implicit representations do not afford: the ability to support dynamic scene updates while preserving semantic consistency. In a neural radiance field, semantic information is implicitly tied to the field's parameters, making it a global, non-discardable component. In 3DGS, however, semantic features are spatially localized attributes, enabling efficient graph-based analysis, such as semantic-aware pruning or targeted updates, and targeted manipulation—a property of significant value in downstream applications requiring compositional editing.

Intriguingly, this rich interaction between the geometric explicit rendering property and semantic abstraction has spawned early explorations into hierarchical semantic decompositions. For object-centric tasks, feature embedding at the primitive level may prove semantically too fine-grained, absorbing intra-object texture variations rather than object identity. A growing zone in the literature therefore argues for spatialization that can identify and preserve the compositionality of objects: part-level awareness using segment-anything masks distilled into a semantic-aware set of shared features, offering a balance between detail and generalizability. This cross-granularity adaptation modulates 3DGS from a shared-scene aesthetic builder to a multi-modal editable index—where semantic abstraction dimensions, texture-like saliency, object boundaries, and instance identifiers coexist and can be queried orthogonally. Future research should continue to move towards **feature-centrism rather than primitive-centrism**, interpreting whether feature refinement and sampling should be performed as a distant layer rather than being tied to individual primitive attributes. The potential for integration with generalized primitive types and hybrid representations is apparent, but realizing these possibilities will require further investigation into the semantic representability of diverse primitive structures, and into transformer- or graph-based aggregation modules that dynamically compose primitives into task-relevant semantic units in real-time.

6.2 Spatial and Appearance Editing Techniques

The editing of 3D Gaussian Splatting (3DGS) scenes is fundamentally empowered by the representation's explicit, primitive-based nature, which establishes a direct correspondence between discrete scene elements and their rendered pixels. This inherent disentanglement—a direct consequence of the primitive-centric paradigm established in the preceding discussion of semantic embedding—supports a broad spectrum of manipulation techniques, ranging from precise spatial operations on localized primitives to holistic, text-driven appearance modifications of an entire object or scene. The discussion bifurcates into two primary, yet intersecting, paradigms: explicit spatial and appearance editing that leverages the geometric and mask-based constraints introduced earlier, and generative, text-driven editing that relies on diffusion priors.

The first paradigm focuses on direct manipulation, exploiting the inherent ease of transforming primitives like the 2D Gaussian surfels from 2D Gaussian Splatting (2DGS) [6]. This explicit nature permits significant dimensionality reduction in the primitive count while simultaneously decoupling geometry from appearance for editing purposes—a complementary capability to the semantic feature fields described in the preceding subsection. For example, neural shell textures allow this decoupling at a global level, meaning that a user can edit the texture on a shell without compromising the underlying geometric fidelity, requiring far fewer primitives. In the same vein, the per-primitive texturing methods like GStex [7] and SVGS [83] allow for simultaneous editing of both 3D asset positions and their local surface properties, such as color or geometry. When the scene topology is simple, a single textured Gaussian can capture complex appearance details, enabling high-fidelity rendering and localized re-texturing or color grading without global optimization [26]. Such content-based texturing ensures that appearance changes are inherently constrained to the designated region, preserving multi-view consistency—a critical requirement that parallels the view-invariance challenges addressed in semantic feature distillation.

Moving towards open-ended editing, the transition to the most advanced paradigm—physical-based relighting and appearance editing—confronts the boundary that geometry editing often results in geometry-appearance fusion. An engineered strategy to overcome this is the design of a decoder that, rather than explicitly incarnating the scene geometry within its parametrization, decomposes appearance information such that it uses geometry only as auxiliary conditioning, thereby categorizing the local properties without altering the underlying structure. Comprehensive methods are proposed to ensure color integrity; most notably, some methods that adopt a photorealism-first architecture [25] inject physically inspired colored-pipelines that the network interprets in a plug-and-play fashion to generate greater control. These decoding techniques—geometry opacities [70] or normal shading—implicitly ensure that any texture or appearance change is applied only to the relevant shader component, leaving geometry untouched. For global edits in style or material, recent work that modifies the bi-directional reflectance distribution function achieves re-shading and relighting from captured data. For instance, 3iGS (Factorised Tensorial Illumination) [110] and Bidirectional Gaussian Primitives [10] may be used to control a scene's illumination. Consequently, an "editing" technique can be defined as an engine operating on separated geometry and appearance halves, conferring a great advantage for accurate relighting and material substitution.

The inherent simplicity in geometry of the splatted primitives remains a core challenge, even when creating novel appearances. The fixed, symmetrical geometry of a Gaussian fundamentally restricts the representational capacity for view-dependent effects, requiring localized feature vectors to compensate for the simplicity of the base shape. In response, the research community is actively exploring alternative appearance representations. Effective methods like Spec-Gaussian [80] and Spherical Voronoi [85] explain the limits of spherical harmonics in representing high-frequency specular effects and provide more powerful, view-dependent appearance encodings. Similarly, the work on SG-Splatting [111] uses Spherical Gaussians in place of third-degree spherical harmonics to reduce parameters and accelerate rendering. To go further, advancing strategies that allocate the primitive count and complexity strategically are crucial. By solving the issue of reducing the memory footprint of 3D Gaussian Splatting [112], a more efficient model can be obtained via a $27\times$ size reduction without losing quality.

As we transition toward the deformation and animation techniques detailed in the next subsection, the editing capabilities described here serve as a crucial foundation. While semantic editing provides the ability to identify and isolate objects, explicit spatial editing manually adjusts their placements and appearances, and the deformation methods will subsequently introduce a higher level of control over the spatio-temporal configuration of these primitives. The explicit separation of geometry and appearance enables seamless binding of the edited structure to a deformable scaffold, as will be discussed, and the primitive-level localization retains the fine granularity needed to preserve editing fidelity.

The field is converging on approaches that address spatial direct manipulation as subsidiary to appearance and physical editing. The intersection of these constraints—semantic, geometric, and physical—offers abundant opportunities for future innovation, envisioning a future where 3D editing is as intuitive as image editing in 2D, yet fully anchored in 3D.

6.3 Deformation, Animation, and Control Structures

The deformation and animation of 3D Gaussian Splatting (3DGS) scenes represent a critical bridge between photorealistic reconstruction and interactive manipulation, addressing the fundamental requirement of binding a high-fidelity surrogate to coarser, user-controllable spatial structures. This capability transforms static reconstructions into editable assets that can be seamlessly integrated into traditional computer graphics pipelines for editing, rigging, physics-based interaction, and real-time animation while preserving rendering fidelity. The research landscape in this domain can be organized around a core tension: how to establish a correspondence between millions of unstructured Gaussian primitives and a manageable set of control handles, with approaches differing in their choice of control structure—meshes, cages, skeletons, or deformable grids—and the mathematical formalism that governs the mapping.

The predominant paradigm leverages high-quality mesh anchors to couple Gaussians to a surface representation. In this setting, each Gaussian is robustly associated with its nearest triangle face using spatial distance metrics, and deformation is propagated via the relative positions of the Gaussians, thereby achieving edge-preserving animation through well-defined coarse-to-dense coordinate mappings. This dense binding strategy, as implemented in Phys3DGS, supports real-time rendering while enabling physically-based relighting and intuitive object-level editing [113]. Similarly, approaches that explicitly model specular surfaces with per-primitive texture mapping bind Gaussians to surface parameterization to facilitate improved appearance modeling [9]. The accuracy of the mesh binding is further refined by hybrid representations that integrate discrete Signed Distance Fields (SDFs) with the primitives, acting as a field to improve geometry and material decomposition [114]. However, mesh-based approaches face the inherent challenge of high-quality mesh extraction itself; the mesh guides the deformation, but a poor initial mesh can severely limit the Gaussians' fidelity to the actual geometry.

In contrast to dense mesh binding, sparse-controlled and cage-based methods provide a lighter-weight deformation mechanism, decoupling the editor's interaction from the scene's geometric complexity. Sparse control points compute smooth deformation fields that warp a larger number of Gaussians, adhering to an as-rigid-as-possible (ARAP) constraint to maintain local shape integrity during editing. Such frameworks support non-rigid body deformation—including bending, stretching, and twisting—by deriving a dense momentum field from a sparse user-specified set, moving the problem away from per-primitive assignment. A recent work on cage-based setups leverages an explicit, learnable control structure to drive dynamic scale and orientation with a three-dimensional deformation field, crucially maintaining the covariance matrix updates [115]. While these methods excel at user-friendliness for sculpting, a limitation lies in the sparsity of the control constraints, which can lead to geometric errors in fine-grained, elongated details.

A third major direction employs skeleton- and proxy-based control structures to parameterize deformations for articulated objects, especially human bodies. Here, Gaussian primitives are associated with control points via skinning weights, and the deformation is directly determined by the skeleton's joint rotations, permitting automatic coupling of Gaussians to surface meshes for physically consistent avatar animation. This binding is essential for a unified rendering and animation pipeline, where the Gaussians and another proxy geometry are fed into a joint renderer, allowing real-time interaction. In this same group, a consistent underlying approach is to use a mesh with rigging functions as the driver, which is crucial for robotic avatars, where the Gaussian model must react in real-time to poses while maintaining high-fidelity details [116]. The trade-offs here manifest in the reliance on proxy mesh quality: while a coarse proxy enables large topology changes (e.g., a skeleton deforming a human limb), secondary effects such as cloth wrinkles often fail to align with the more detailed Gaussian distribution, creating a visual discontinuity.

Looking at the diversity of solutions, the major technical axis is the binding density of the control structure to the Gaussian primitives, whether through rigid meshes, sparse cages, or skeletons; and a conclusive trend is emerging towards hybrid and optimized control structures, wherein the deformation is decoupled from direct photometric error and constrained by topology-based regularizers. Each strategy demonstrates distinct trade-offs: mesh structures provide the highest local shape fidelity but struggle with topology change, while cage and skeleton-based methods offer global addressing but are limited in local geometric precision. Future research is likely to focus on learning-based approaches for adaptive control structure generation from the Gaussian representation itself, potentially eliminating the manual authoring of binding regions. This would provide a truly generalizable framework for interactive animation, uniting the scalability of sparse control with the local accuracy of mesh-based, fully paired surface-aware approaches that are fast becoming the standard for 4D content creation.

6.4 Scene Generation and Content Synthesis with Generative Priors**

Building upon the deformation and animation frameworks that establish spatio-temporal control over Gaussian primitives, the synthesis of static assets and dynamic scenes from user-provided prompts represents the next frontier: the automatic generation of these deformable and editable representations. This paradigm shift in 3D content creation positions 3D Gaussian Splatting (3DGS) as a powerful target representation for generative pipelines. Unlike implicit neural fields, the explicit, point-based nature of Gaussian primitives is uniquely compatible with the output of modern generative models, enabling direct optimization, real-time rendering, and crucially, the seamless integration of the deformation and binding strategies discussed previously. These methodologies integrate foundational 2D and 3D diffusion models with the differentiable rasterizer to lift concepts from text or image prompts into explicit, high-fidelity scenes, effectively transforming generative priors into structured radiometric point clouds.

A dominant strategy in this domain is multi-stage generation guided by score distillation sampling, which aligns 3D representations with 2D diffusion priors. The objective of such pipelines is to optimize the Gaussian primitives so that their rendering from random view-points is consistently classified by a pre-trained 2D diffusion model as a highly plausible image. In this paradigm, the diffusion model acts as a learned, operational constraint rather than a source of direct ground truth, enforcing structural and probabilistic appearance consistency of the generated object. However, this optimization is susceptible to mode collapse, deriving unrealistic geometry from a single image—often manifesting as a variant of the Janus problem. To mitigate this instability, recent approaches have introduced geometric priors that resonate with the coarse-to-fine binding philosophies of the deformation section. For instance, generating per-view RGBD images via diffusion and lifting them into a 3D Gaussian scene before refinement, or initializing sparse points from a single image's depth estimate, anchors the optimization in a structured, albeit noisy, scaffold. This process dramatically reduces the search space and yields higher-fidelity assets, effectively pre-binding the Gaussian primitives to a rough spatial configuration.

Moving to text-conditioned generation, robust control over both geometry and appearance requires an intermediate, structure-aware phase to decouple semantics from pixel-space optimization. Here, the use of scaffold embeddings—originating from user-defined layouts or LLM-driven scene scripts—converts complex queries into per-object semantic masks. This decomposition conditions the Gaussian optimization graph on fine-grained spatial regions, akin to the local control strategies of sparse cages but applied at the semantic level. By breaking down a complex scene into its constituent assets, the generative model can inject specific object priors without cross-contamination, ensuring instance-level geometric registration. As such, these per-primitive semantic anchors become the generative analogue of the control skeletons: they provide a sparse, editable constraint set that governs the volumetric arrangement of the generated primitives, aligning with the need for adaptive control structure highlighted at the closing of the previous section.

Dynamic 4D generation extends this synthesis to the temporally variable domain, combining the static anchors with the deformation techniques previously discussed. These pipelines typically require an iterative refinement loop: starting from an initial 3D asset bound to a motion template or monocular video, the system propagates small spatio-temporal increments, preserving both spatial and temporal consistency. By optimizing a joint diffusion and motion constraint—where the deformation field is regularized through a 3DGS surface—the generated dynamic scenes remain coherent and free of accumulative errors, ensuring that the Gaussians adhere to their assigned temporal control.

In conclusion, the confluence of 3DGS, score distillation, and conditional diffusions is forging a unified and coherent pathway for dynamic scene creation. The future trajectory will blend the local adaptability of the deformation field (learned control meshes or skeleton proxies) with the global generalization of the generative model, thereby replacing the manual binding processes described in previous subsections with generation-driven, automated acquisition of complete, rigged assets from observations. This opens been directly along the steepest gradient towards the end of our survey, demonstrating that 3DGS is not only a fully editable, but a generative, animated, and interactive workspace—making synthesis, editing, and rendering but facets of a single, unified pipeline rather than distinct, sequential authoring steps.

7 Real-World Applications and Advanced Interaction

7.1 Robotics, Autonomous Navigation, and SLAM

The integration of 3D Gaussian Splatting (3DGS) into robotics represents a paradigm shift from offline visual reconstruction to online, closed-loop perception. The explicit, point-based nature of the Gaussian representation—which provides immediate access to scene geometry and appearance—combined with its differentiable tile-based rasterizer, makes it uniquely suited for the core autonomy stack of mapping, localization, and navigation. By transitioning from the slow per-ray sampling of implicit neural radiance fields (NeRFs) [20, 1] to a fast primitive-splatting pipeline, 3DGS allows for a unified representation that can be optimized in real time to provide the dense, metric scene understanding required for safe and efficient operation in complex, unstructured environments.

Real-time dense SLAM and state estimation is the most prominent application area. Traditional SLAM systems rely on sparse feature maps or dense depth fields, which are either insufficient for collision avoidance or computationally expensive. 3DGS-based systems have been developed to address these shortcomings, though the original algorithm's assumption of a pre-computed structure-from-motion (SfM) point cloud presents a fundamental bottleneck. To overcome this, modern frameworks [1, 4] have introduced novel initialization strategies that operate directly on monocular, stereo, or RGB-D video streams. For instance, a crucial architectural adaptation involves decoupling the pose update from the map density control process to avoid cascading errors during joint optimization. While the anchor-driven deformable and compressed methods, such as those proposed for dynamic scenes, demonstrate the necessity of compactness for memory-limited robots [117, 118], the direct use of geometric constraints derived from depth or range data is critical. The formulation allows the map to be updated incrementally by extracting local gradient signals from photometric loss and structural similarity, enabling "frame-to-model" tracking that is robust under aggressive motion. The work on refined representations [101] that introduces static constraints and motion consistency suggests that mapping in dynamic environments requires explicit handling from a representation perspective. However, the impact of these innovations in tracking state loops is evident: the explicit geometry of the Gaussians facilitates direct ray-ellipsoid intersection for differentiable depth rendering [16], which in turn provides the metric scale and geometric supervision needed to prevent drift in real-world autonomous platforms.

Beyond tracking, **robust localization and map-based pose tracking** leverage the high-fidelity novel view synthesis of a pre-built Gaussian model as a reference environment. The ability to render photorealistic views from arbitrary camera poses allows for direct image registration, where the pose is estimated by minimizing the photometric residual between the observed and rendered images. To increase robustness under varying lighting or seasonal changes, authors have augmented these models with LiDAR data, utilizing the geometric accuracy of depth measurements to refine the Gaussian shapes. Given the sensitivity of 3DGS to optimization failure and local minima ("blur traps") [3], foundational works [17] have proposed plug-and-play network modules, which densify the original sparse SfM point clouds to provide a more reliable geometric structure. This allows for filtering unstable or redundant points and improves the robustness of dense reconstruction in environments with large or occlusion-prone scenes. A comprehensive framework–such as that used for large-scale road networks in autonomous driving [14]–demonstrates the necessity of temporal caching and block splitting to keep memory bounded, generalizing effectively across large-scale and complex environments.

Perception and navigation in complex environments extend beyond visual fidelity to higher-level tasks such as scene understanding and semantic interpretation. Here, the real-time nature of 3DGS can be leveraged for tasks such as collision avoidance or semantic segmentation, without the need for expensive multi-path calculations [119]. In more semantic applications, the integration of 3DGS into a full feedback loop is frequently required for safe operation. For example, adding language features to the semantic features allows for language-guided navigation in everyday environments ("find the red box behind the table") [12]. This reduces search spaces and extends foundational prior knowledge into the model. However, robustness requires posing a maintenance challenge. Every dynamic object (e.g., moving vehicles, pedestrians) violates the original assumptions, presenting a challenge for the original formulation. Frameworks from dynamic scene reconstruction, such as hybrid 3D-4D representations [119] that adaptively allocate 3D Gaussians to static areas while retaining 4D primitives for moving parts, have shown high efficiency in tracking receptivity and reducing drift through their explicit temporal modeling.

Despite the advances, several critical challenges remain, specifically when addressing large-baseline scenes and the arbitrariness of object-centric map management. The majority of these methods do not yet simplify complex tasks such as re-optimization and relocalization for shape approximation processes. The transition from existing systems suggests a forward-looking future that will depend on principles beyond appearance, meaning that they should more comprehensively leverage 3DGS geometry information [120]. In addition, movable agents are increasingly integrated into tasks such as editing and simulation [1, 119]. The trend in aligning with physics-based simulation [120] indicates that robot action will be brought into the accurate reconstruction, creating a fully self-contained loop where the robot learns dynamics and visual appearance its environment. The broad application in GPS-denied environments open possibilities for large-scale, persistent, and collaborative "digital twins" that are maintained under a small robot fleet. As 3DGS continues to improve robustness and memory efficiency with adaptive densification and learning-based extraction (e.g., [17, 121]), the compounding [representation will very likely become an engine for, not just a map in real time but also understand the physical constraints and semantic code that make autonomy. This forms a mutually beneficial relationship between synthesis and autonomy, promising a future where 3DGS serves as a central backbone for robotic perception and interaction.

7.2 Large-Scale and Aerial-to-Ground Scene Modeling

The transition from object-centric reconstruction to unbounded urban environments represents one of the most demanding scalability challenges for 3D Gaussian Splatting (3DGS). While the robotics applications discussed in the previous section prioritize real-time, closed-loop operation with tight latency and memory budgets, city-scale reconstruction introduces a complementary yet equally stringent set of constraints: environments must scale to millions of primitives, accommodate extreme dynamic range in viewing distances, and reconcile significant appearance variations arising from multi-sensor capture platforms. These urban scenes are typically reconstructed from heterogeneous sources—aerial imagery provides broad, top-down contextual coverage, while street-level captures deliver close-range, high-frequency detail; naively fusing these modalities produces ghosting artifacts and geometric inconsistencies. Recent progress has therefore coalesced around three interrelated axes that directly inform the deployment challenges—such as memory-limited rendering and streaming—that will be examined in the following subsection on mixed reality and embedded systems: hierarchical spatial partitioning, appearance harmonization, and modality-aware geometric fusion.

Memory and computation scalability in the original 3DGS framework is fundamentally constrained by its dense, unstructured primitive set and global optimization schedule, which fails when a scene exhibits millions of Gaussians—a limitation that becomes even more acute when these scenes must later be deployed on bandwidth-limited mobile platforms. A dominant strategy to address this is block-wise decomposition, dividing the unbounded space into manageable sub-regions that can be optimized independently via distributed multi-GPU pipelines. While straightforward, static subdivision suffers from combinatorial explosion in inter-block visibility and load-balancing. More robust solutions integrate level-of-detail (LOD) hierarchies with partitioning, where a global coarse field captures district-level context while finer local blocks are activated based on camera proximity, directly matching the system to a controllable memory budget. This dual-primitive strategy enables real-time interaction by effectively performing view-dependent culling at multiple spatial granularities—a capability that naturally aligns with the streaming and progressive-loading needs of virtual production and MR pipelines discussed in the following section. The density-stratification patterns observed in converged solutions—where dense regions exhibit geometry-correlated parameters amenable to render-free prediction, while sparse regions show systematic failure across architectures [122]—underscore the importance of such adaptive, geometry-aware allocation, particularly for guiding the primitive-compression and pruning strategies that are essential for on-device rendering.

The integration of aerial and ground data introduces geometric and photometric discrepancies that are fundamentally misaligned—aerial views offer top-down coverage with low occlusion, while ground views capture facades and street-level details. Directly optimizing a model from both induces blur and ghosting. Advanced pipelines address this challenge via various strategies: coarse-to-fine training strategies eliminate floaters caused by overfitting in real-world scenes [80], and MVS-guided initializations combined with optimization strategies refine Gaussian placements [29]. Furthermore, dynamic allocation of Gaussians based on localized scene complexity, guided by predicted cardinalities rather than fixed per-area budgets, shows promise for efficiently allocating primitives in heterogeneous scenes [123]. These geometric refinements ultimately produce cleaner primitive sets that are not only more faithful but also far more compressible—a property that directly informs the quantization, pruning, and distillation techniques developed for embedded deployment in the following subsection.

The other prominent complication in aerial-to-ground urban reconstruction is the appearance disagreement between capture modalities—sun angle changes as a function of capture moment, coupled with vastly different camera responses between aerial sensors and ground car-mounted rigs. This leads to temporally and sensor-specific color variations that break multi-view consistency; naive appearance adaptation often results in visible blockiness and seams. Solutions analogous to image-level translation techniques [86] are needed, together with coarse-to-fine training and spectral harmonization, to achieve visual continuity. Such strategies serve as a precursor to rigorous photometric normalization, ensuring that the model learns a clean, physically consistent layer that does not memorize transient appearance artifacts.

The topic of scalability is gradually shifting from simply loading large scenes toward *ensuring that densified approaches support richer appearance and novel view synthesis for low-power mobile and embedded computing*. This shift is driven by the reliance on LOD strategies to control complexity in a way that transparent and GPU-compatible primitives can be handled, directly bridging to the sort-free and order-independent rendering methods employed for multi-device display in the next subsection. Future directions are likely to coalesce around learning-based sparse optimizers (e.g., gradient-guided density control and forgetful-geometry regularization [29, 112]) and batch-based training routines that amortize toward memory-limited edge rendering. Yet key challenges remain clearly in handling inherent appearance sensitivity, and in guaranteeing consistency, robustness, and quality across heterogeneous platforms and capture conditions—especially when these large-scale urban models must later be deployed within the interactive, digital-twin, and virtual-production pipelines discussed next.

7.3 Virtual Production, Mixed Reality, and Embedded Systems

The deployment of 3D Gaussian Splatting (3DGS) in virtual production, mixed reality (MR), and embedded systems represents a paradigm shift from offline, desktop-centric rendering to latency-critical, bandwidth-limited, and compute-constrained environments. Unlike traditional mesh pipelines, the explicit, point-based nature of 3DGS offers a unique opportunity for high-fidelity novel view synthesis, yet its practical viability hinges on overcoming fundamental barriers: the computational cost of sorting operations, the sheer memory footprint of primitive sets, and the synchronization challenges in distributed workflows. The transition to these demanding domains requires a holistic re-engineering of the entire pipeline, from primitive representation to hardware abstraction.

A primary bottleneck for embedded and mobile rendering is the reliance on depth-sorting for alpha compositing, a process that is difficult to parallelize efficiently and introduces significant overhead on architectures with limited compute and memory bandwidth. This has driven a critical line of research toward **order-independent rendering paradigms**. The inherent non-commutativity of alpha blending necessitates view-dependent sorting, which [39] identifies as a key performance constraint on mobile platforms. They propose a "Weighted Sum Rendering" approximation that bypasses sorting entirely, achieving speedups without sacrificing competitive image quality by optimizing a generalized formulation amenable to gradient-based refinement. Building on this, [124] introduces a dual-hierarchy framework using cell proxies to further accelerate rendering, and couples this with a physically-inspired weighted sum technique that mitigates both "popping" and "transparency" artifacts typically associated with order-independent approximations. This approach achieves order-of-magnitude speedups with substantial reductions in rendering overhead. Alternatively, the removal of sorting can be achieved through principled stochastic or regression-based approaches. [115] leverages an unbiased Monte Carlo estimator of the volume rendering equation, allowing for accurate 3D blending of overlapping Gaussians and, critically, providing a direct and continuous trade-off between computational cost and visual fidelity by varying the number of samples per pixel—a feature that showcases the differences with standard rasterization. Further, advanced methods like [47] employ a lightweight multilayer perceptron to approximate transmittance, enabling alpha blending without explicit sorting, a strategy that is integral to unlocking high-performance on power-limited embedded systems.

A second fundamental challenge lies in the inflexibility and memory large footprint of primitive representation, especially when targeting real-time performance on mobile devices. The typical need of sorting-based forward rendering requires high memory and memory access, and the underlying representation is difficult to compress. To achieve real-time performance in resource-constrained environments, addressing these bottlenecks is crucial. Recent efficiency advances tackle these issues. For example, methods that reduce the memory footprint by compressing the 3D Gaussian representation through contributions-based pruning, vector quantization, and first-order spherical harmonics distillation, achieving significant model size reduction while maintaining high quality [125]. The work in [126] addresses the GPU warp divergence and bounding-box inefficiencies by performing a "warp-coherent" rendering paradigm. By recognizing the spatial association between Gaussian kernels and SIMT execution boundaries, they organize entire warp-level structure and achieve significant speed-ups. Similar hierarchical designs in [88] partition the rasterization into a "tolerant hierarchical" structure, decoupling "data partitioning" from the rendering to reduce the redundant bottlenecks in preprocess. Notably, the [127] method is designed to be a resource-efficient framework that can, for itself, preserve the memory advantages of a precision-optimized rasterizer, and it is possible to adapt to these mobile constraints.

In virtual production pipelines, the integration of live Gaussian assets into established render engines (e.g., Unity, Unreal) is a practical impediment. To bypass interactive compatibility issues, [128] uses Nvidia's interprocess communication (IPC) APIs to establish a GPU-side data-sharing mechanism. This decouples the GS renderer from the final display and allows for the streaming of visualization results into arbitrary software clients, such as Blender or Unreal, enabling a broad range of applications (e.g., cut scenarios, engine-agnostic tool). This flexibility is essential for the studio, where the blending of many traditional and novel assets is required.

Finally, connectivity is central to the pipeline. As the number of assets grow to form the homascale 3D world, the memory limitations and high bandwidth requirements are an obvious challenge. High-quality and large-scale scenes require streaming and progressive loading. To accommodate a variety of display devices, it is necessary to devise adaptive and level-of-detail strategies that can scale **3D Gaussian primitives**. One viable path is the adaptation of an algorithmic-primitive representation. For instance, to achieve video-streaming deployment of an `adaptive primitive` such as scattered decomposition methods in [22] provide a pathway—by structuring primitives with truncatable coefficients in a single trained panel, we can control the model's rendering complexity to stream at varying rates (LOD), making them a naturally `scalable` and compressible data solution for minimizing network bandwidth and memory [50].

Taken together, these interdisciplinary developments show a clear trajectory: the after-deployment of 3DGS to MR and mixed reality is in a constant tug-of-war between the demand for maximum visual fidelity and the flexibility to balance between several computational and memory budgets. While eliminating sorting (either by stochastic or regression-based transmittance estimation [115], [47]) is an advantageous avenue, they fundamentally alter the mathematical interpretation of the rendering, demanding attention to the "energy" that is being approximated (see theories in [33], [34]). While [39] indicates that we cannot exactly replace the physically meaningful volumetric evaluation with faster strategies, the recent surge of specialized techniques demonstrates that approximations can often be more scalable and well-supported than the previous work. Novel future research will further breakdown the `multi- architectural` boundaries, e.g., by jointly co-designing the algorithms with the parallel hardware, or by creating task-specific models of the rendering operation, as this joint design is necessary to fully unlock the potential of 3DGS in interactive, digital-twin, and high-fidelity virtual-production pipelines for real-time users.

7.4 Physics-Integrated Simulation and Digital Twins

The convergence of photorealistic scene reconstruction and physically plausible interaction represents one of the most consequential frontiers for 3D Gaussian Splatting (3DGS). Building upon the scalability and appearance-handling strategies discussed for urban-scale reconstruction, and extending the optimization robustness themes that follow, physics-integrated frameworks transform Gaussian primitives into digital twins that both look real and respond to applied forces, heat, or collisions. This capability stems directly from the explicit, particle-like nature of Gaussian splats, which naturally intersect with established computational mechanics pipelines such as the Material Point Method (MPM), Finite Element Method (FEM), and Position-Based Dynamics (PBD).

A foundational challenge in this domain lies in the geometry abstraction required to bridge the primitive-based visual representation with continuum mechanics solvers. While Gaussian splats excel at photometric fidelity, they are not intrinsically compatible with the Lagrangian particles or tetrahedral meshes typically consumed by physics engines. Solutions emerge through sophisticated mechanisms that convert radiance-carrying primitives into unified physical particle sets. These abstracted assets—be they deformed Gaussian splats, standard CG meshes, or fluid particles—are re-expressed in a solver-agnostic kernel, thereby facilitating scene-level interactions among heterogeneous assets. This abstraction layer is not merely a format conversion; it critically must preserve mass, momentum, and volume properties to ensure that the ensuing simulation is physically plausible rather than merely visually satisfying.

Within this integrated framework, a second key area of technical contribution centers on material-aware dynamics, fracture, and thermo-mechanical coupling. Techniques discretize explicit Gaussian geometries to model brittle fracture [57], and also extend the formulation to use multi-material bindings that are essential for structural analysis. Crucially, these systems must address how temperature propagates within the radiance field. Through parameterizing a heat field into physical constraints, one can align and schedule heat transfer with mechanical deformation—for instance, simulating melting, softening, or thermal expansion—thus enabling digital twins that exhibit complex thermo-mechanical behaviors, such as heat-induced failure or welding. The integration of such physics engines has also been demonstrated to calibrate structural inconsistencies that arise from non-rigid inertial errors, offering stronger guarantees than property-based visual simulation alone.

A third technical area of this research thread is the anchoring of visual primitives within a prediction-correction loop. Merely simulating a mesh does not generate an updated, photorealistic rendering; the two representations must be synchronized. This is achieved by binding a set of 3D Gaussians to a coarser, position-based dynamics mesh or rigid body controller. The anchoring strategy ensures spatial cohesion, and continuous visual feedback from the Gaussian renderer provides photometric corrections to the physical state. This creates a closed-loop design where Gaussian splats remain the sole renderable representation, while position-based dynamics meshes provide the structure-agnostic trajectories for the visual primitives. Such an approach not only maintains physical stability but also permits the manipulation of scene materials via the constraints defined at the solver level, essentially enabling real-time global editing of a scene's physical properties.

Compared to volumetric neural fields, which struggle to provide stable gradients for physical contact or coherent particle tracking, the explicit structure of 3DGS naturally stabilizes contact and constraint handling without needing high-frequency re-sampling. Methods such as [91] can be co-opted to provide primitives that respond to a frequency-aware learning rate—essential for capturing the high-frequency details without triggering an inefficient proliferation of small Gaussians. Moreover, techniques like [57] offer the crucial benefit of formally reducing the number of points in a physics simulation without sacrificing the final rendering quality, thereby drastically cutting computation and improving efficiency and democratizing the use of GS for real-time interactive digital twins.

Looking forward, the coupling of physics-derived gradients with the visual-loss requires a unified optimization that is not yet fully developed—the two sub-fields still largely rely on guided heuristics in the interaction. However, the trajectory converges toward the central goal of building digital twins that faithfully mirror both the composition of natural and artificial structures and their dynamic response to the complexity of the real world. Future work should target a **adaptive co-design** of density control and physical constraints that can be simultaneously templated across large datasets, enabling seamless embodiment of unforeseen phenomena without extensive manual calibration [53, 129]. This pursuit reinforces the overarching movement toward a unified, reliability-centric view of 3DGS as a foundational world-model primitive—bridging from the multimodal and robustness challenges discussed next to a physically grounded, safety-aware interactive world model.

7.5 Emergent Domains: Multimodal Sensing, Generation, and Safety

3D Gaussian Splatting is rapidly transcending its origins in RGB novel-view synthesis, emerging as a versatile core representation for a broader class of multimodal sensing, generative content creation, and safety-critical systems. This expansion is propelled by the explicit, point-based nature of Gaussians, which offers a unique interface for integrating non-visual sensor data, serving as a powerful output representation for generative models, and enabling rigorous robustness analysis. This subsection explores these emergent domains, highlighting the technical innovations that facilitate this diversification and evaluating the associated opportunities and challenges.

In the context of multimodal sensing, the conventional dependence of 3DGS on Structure-from-Motion (SfM) for initialization presents a significant bottleneck for real-world, multi-sensor deployment [95]. While learning-based initialization strategies, such as those leveraging 3D foundation models to jointly estimate poses and Gaussian primitives, show promise for fast reconstruction [63], they often rely on well-conditioned visual input. The integration of active sensors like LiDAR offers a more principled solution. By jointly optimizing camera poses and scene geometry from coarsely-posed images and noisy LiDAR point clouds, robust pipelines can be established without the need for SfM [98]. This approach highlights a key trend: the representation's explicit geometry can serve as a fusion medium, absorbing depth information to constrain the optimization problem in ways that are challenging for purely implicit methods. This robustness to sensor noise and pose error is a critical advantage, aligning Gaussian maps with the demands of acquisition systems with limited views [100, 130, 32].

The frontier of robust and efficient optimization is being advanced by addressing the intrinsic initialization bottlenecks of 3DGS. A common theme involves developing strategies to reduce reliance on SfM-derived point clouds. For instance, some approaches propose progressive frameworks that optimize 3DGS during image capture, updating poses and Gaussians on-the-fly as new images arrive [97]. Others focus on eliminating the densification process altogether by generating a dense initialization from triangulated pixels across input images, which reduces the optimization path and enhances detail in high-frequency regions [32]. To mitigate the issues of noisy or sparse initialization, some methods incorporate spatial-aware denoising strategies—formulating optimization as a primitive denoising process that considers both position and spatial structure [96]—or use structure-aware distribution and region-prioritized optimization to encourage balanced structural coverage and faster convergence [60]. Furthermore, exploring strategies to escape local optima during training is essential, as the optimization landscape of Gaussian primitives is prone to getting trapped in "blur traps" [64].

The robustness of the GS representation itself is a growing area of concern with significant practical impact. When engaged with navigation and perception, the fidelity and consistency of the Gaussian map becomes paramount. To achieve this, some works emphasize the need for careful initialization strategies that support generalization across diverse real-world conditions [95, 93, 99]. As an alternative to SfM, certain methods leverage pre-trained 3D foundation models to establish a strong starting point for optimization from limited input views, controling training time and computational cost [63]. Work also suggests that densification strategies increased during training can improve robustness to noise in dynamic scenes [38]. Moreover, the robustness of the optimization process to various levels of noisy inputs and initialization quality—from random to coarse—is a testament to ongoing progress in a field that seeks to guarantee consistent and high-quality model generation [92, 64]. The quest for reliable and safe application of 3DGS in mission-critical domains extends beyond modeling pipelines.

Emerging frontiers also include safety-critical analysis and adversarial robustness—a domain where the explicit nature of Gaussians enables more thorough threat assessment. Monte Carlo simulations can be used to evaluate the robustness of neural, but for 3DGS, the tolerance of the model to **noisy, sparse, or incorrectly posed images** is the most critical scenario [98]. Reliable analysis of the robustness of 3DGS under sparse-view settings is facilitated by initializations designed for generalization, as evaluated by specific point-cloud priors and frequency-aware SfM [94]. When such robustness, methods that allow the system to "know what it does not expect" can provide a critical safety buffer. This necessitates research into **confidence-bounded** rendering and the formal certification of the model under bounded perturbations remains a hard and uncharted research area within the 3DGS community, requiring new statistical and adversarial reasoning.

In summary, the frontiers explored in this subsection indicate a significant maturation of 3D Gaussian Splatting as a multimedia-centric 3D representation. More direction goes beyond the traditional pixel-wise objectives to include a set of new sensor modalities, generative priors, and safety-aware optimization landscapes. The other conditional trend is toward decoupling the quality of the representation from the quality of the raw input and its initialization. This yields improved robustness to a range of degradation and opens new avenues for a new class of systems in which high-fidelity major reconstruction can be the product of emergent and multimodal model integration. Future work here will center on developing a unified framework for uncertainty-aware, physics-coupled, and generative 3DGS that is also safe by design, enabling an era of world models that are not only photorealistic but also physical, reliable, and robust in the loop.

8 Conclusion

This survey has charted the rapid ascent of 3D Gaussian Splatting from a novel rendering technique to a foundational paradigm for explicit, real-time scene representation. The current landscape is defined by a remarkable convergence of capabilities: the core 3DGS algorithm achieves photorealistic rendering at unmatched speeds by unifying high-fidelity spatial representation with a differentiable rasterization pipeline [2]. This has precipitated a proliferation of derivative works that have expanded its utility far beyond novel view synthesis, transforming 3DGS into a versatile and foundational tool for 3D vision and graphics [4]. The explicit nature of the primitive-based representation facilitates precise editing, dynamic scene reconstruction, and semantic understanding, unlocking applications in robotics, virtual production, and advanced interaction [3, 1, 18].

Despite these strides, several significant bottlenecks have also emerged. A major limitation is the substantial memory footprint of a trained model, which requires millions of learnable primitives and excessive storage, hindering deployment on resource-constrained devices [112, 1]. This has spurred a rich body of work addressing the scalability and efficiency of the representation [15, 118, 131]. Furthermore, critical dependencies on initial observation, such as Structure-from-Motion (SfM) point clouds, can limit generalization and practicality, necessitating more robust initialization methods or flexible modeling approaches [17, 132]. These shortcomings have prompted exploration of new perspectives, including alternative key components [11], understand of the model's approximations [67], and more efficient or accurate representations like textured or 2D Gaussian surfaces [7, 8, 6].

Perhaps the most compelling recent frontier is the integration of 3DGS within dynamic and physics-aware systems. The extension to temporal domains has branched into two paradigms: deformation-based approaches that map from a canonical Gaussian space [133, 102] and direct 4D spatio-temporal representations [19, 134]. While deformation fields can help enforce temporal smoothness, direct 4D primitives offer extreme rendering speeds, yet often face significant storage overhead, motivating the development of hybrid models or optimizations to reduce redundancy [119, 117]. The work in developing digital twins and physically-based simulators is poised to become more unified and impactful [4].

Looking forward, the field is marching towards generalizable and feed-forward solutions. Beyond per-scene optimization, generalizable networks seek to reconstruct complete scenes from sparse views without needing a lengthy per-scene optimization phase [132, 17]. The use of Level-of-Detail (LOD) to handle unbounded, massive-scale scene is less important as a solution to directional, and continuous operations [65]. While the theoretical underpinnings of the splatting and its optimization remain an area of depth that is not explicitly resolved beyond its initial success, the empirical progress is undeniable. The future of 3D Gaussian Splatting lies in more resource-efficient and interactive, becoming the standard solution for digital asset creation and realistic rendering.

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